

A COMPACT COMPLEX THREEFOLD DIFFEOMORPHIC TO THE SIX-SPHERE

STREAMLINING BY CHATGPT SOL 5.6 PRO OF ANTHROPIC'S ORIGINAL PROOF, AS
COMMUNICATED BY LEVENT ALPOGE

ABSTRACT. We construct a compact complex threefold fibred by complex two-tori over $\mathbb{P}^1$. The fibration has one reduced, nonnormal normal crossings fibre and two multiple fibres of multiplicities 3 and 4. We prove that the total space is simply connected and compute its integral cohomology by the Leray spectral sequence. The local calculation at the nonnormal fibre is carried out integrally by nearby cycles. The total space is an integral homology six-sphere and hence is diffeomorphic to the standard S^6 .

1. STATEMENT AND CONVENTIONS

Theorem 1.1. *There is a compact connected complex manifold X of complex dimension three whose underlying smooth manifold is diffeomorphic to the standard six-sphere. Consequently S^6 admits an integrable complex structure.*

The proof presented here is a streamlining by ChatGPT Sol 5.6 PRO of an original proof produced by an AI model internal to Anthropic, as communicated by Levent Alpoge and circulated as [1]; all arguments used for the theorem are included here.

All lattices are written additively. Exterior products are abbreviated by juxtaposition, for example $\gamma u = \gamma \wedge u$. Paths are composed from left to right. All meridians are oriented clockwise; based meridians a_1, a_2, a_0 about p_1, p_2, p_0 , respectively, are chosen so that $a_1 a_2 a_0 = 1$.

2. INTEGRAL MONODROMY AND THE BASE ORBIFOLD

2.1. The lattice representation. Let $V = \mathbb{Z}^4$ with ordered basis (γ, u, w, δ) and let $\Lambda = V^*$ with dual basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$. Evaluation on γ is denoted by the same letter:

$$\gamma(a\hat{\gamma} + b\hat{u} + c\hat{w} + d\hat{\delta}) = a.$$

Set

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For $T \in \text{GL}(V)$ write $A(T) = (T^{-1})^t \in \text{GL}(\Lambda)$, and put $A_j = A(T_j)$.

Proposition 2.1 (Integral linear algebra). *The following statements hold.*

(i) T_1 and T_2 have exact orders 3 and 4, respectively, and

$$T_0 := (T_1 T_2)^{-1} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I + N, \quad N^2 = 0,$$

with

$$N\gamma = Nu = 0, \quad Nw = -u, \quad N\delta = \gamma.$$

(ii) In the dual basis,

$$\begin{aligned} A_1 \hat{\gamma} &= \hat{\gamma} + 6\hat{u} - 6\hat{w} - 2\hat{\delta}, & A_1 \hat{u} &= -\hat{w} + \hat{\delta}, & A_1 \hat{w} &= \hat{u} - \hat{w}, & A_1 \hat{\delta} &= \hat{\delta}, \\ A_2 \hat{\gamma} &= \hat{\gamma} - 6\hat{w} + 3\hat{\delta}, & A_2 \hat{u} &= \hat{w}, & A_2 \hat{w} &= -\hat{u} + \hat{\delta}, & A_2 \hat{\delta} &= \hat{\delta}. \end{aligned}$$

Writing $M_0 = A(T_0)$, one has

$$M_0\widehat{\gamma} = \widehat{\gamma} - \widehat{\delta}, \quad M_0\widehat{u} = \widehat{u} + \widehat{w}, \quad M_0\widehat{w} = \widehat{w}, \quad M_0\widehat{\delta} = \widehat{\delta},$$

and $A_1A_2M_0 = I$.

(iii) Put

$$\Lambda_{\text{tor}} = \langle \widehat{w}, \widehat{\delta} \rangle, \quad \overline{\Lambda} = \Lambda / \Lambda_{\text{tor}}.$$

Then

$$\ker(M_0 - I) = \text{im}(M_0 - I) = \Lambda_{\text{tor}},$$

and the induced map

$$B_0 : \overline{\Lambda} \longrightarrow \Lambda_{\text{tor}}, \quad B_0\overline{\widehat{\gamma}} = -\widehat{\delta}, \quad B_0\overline{\widehat{u}} = \widehat{w},$$

is unimodular.

(iv) The vectors

$$\varepsilon = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \quad \varepsilon' = \widehat{\gamma} + 3\widehat{u} - 3\widehat{w}$$

satisfy

$$\Lambda^{A_1} = \langle \varepsilon, \widehat{\delta} \rangle, \quad \Lambda^{A_2} = \langle \varepsilon', \widehat{\delta} \rangle.$$

We fix the twist vectors

$$v_1 = \varepsilon, \quad v_2 = -\varepsilon'.$$

Thus $\gamma(v_1) = 1$ and $\gamma(v_2) = -1$.

(v) If $G = \langle T_1, T_2 \rangle$, then

$$V^G = \mathbb{Z}\gamma, \quad \sum_{g \in G} \text{im}(g - I) = \langle \gamma, u, w \rangle,$$

while

$$\Lambda^G = \mathbb{Z}\widehat{\delta}, \quad \sum_{g \in G} \text{im}(A(g) - I) = \ker \gamma.$$

In particular the coinvariants V_G and Λ_G are free of rank one. The smallest $\langle A_1, A_2 \rangle$ -invariant subgroup of Λ containing Λ_{tor} is $\ker \gamma$.

(vi) Set

$$q = uw + 6\gamma\delta, \quad \text{vol} = \gamma uw\delta.$$

Then

$$(\bigwedge^k V)^G = \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}\gamma, & k = 1, \\ \mathbb{Z}q, & k = 2, \\ \mathbb{Z}\gamma uw, & k = 3, \\ \mathbb{Z}\text{vol}, & k = 4, \end{cases}$$

and the coinvariants are

$$V_G = \mathbb{Z}[\delta], \quad (\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta], \quad [uw] = 6[\gamma\delta], \quad (\bigwedge^3 V)_G = \mathbb{Z}[uw\delta], \quad (\bigwedge^4 V)_G = \mathbb{Z}[\text{vol}].$$

Finally

$$\begin{aligned} V^{T_0} &= \langle \gamma, u \rangle, \\ (\bigwedge^2 V)^{T_0} &= \langle \gamma u, \gamma\delta, uw, \gamma w - u\delta \rangle, \\ (\bigwedge^3 V)^{T_0} &= \langle \gamma uw, \gamma u\delta \rangle. \end{aligned}$$

All displayed invariant subgroups are saturated, and all displayed coinvariant groups are torsion-free.

Proof. Direct multiplication gives $T_1^3 = T_2^4 = I$ and the displayed formula for T_0 . The formulas for A_1, A_2, M_0 follow by taking inverse transposes. For $\lambda = a\hat{\gamma} + b\hat{u} + c\hat{w} + d\hat{\delta}$,

$$(M_0 - I)\lambda = -a\hat{\delta} + b\hat{w},$$

which proves (iii). Solving $(A_j - I)\lambda = 0$ gives $b = 2a, c = -4a$ for $j = 1$, and $b = 3a, c = -3a$ for $j = 2$, proving (iv).

For (v), solving $T_j x = x$ gives

$$V^{T_1} = \langle \gamma, 2u + w + 3\delta \rangle, \quad V^{T_2} = \langle \gamma, u + w + 2\delta \rangle,$$

whose intersection is $\mathbb{Z}\gamma$. The images of $T_j - I$ contain $u, w, 2\gamma, 3\gamma$, hence generate $\langle \gamma, u, w \rangle$. Dually, the images of $A_j - I$ generate $\langle \hat{u}, \hat{w}, \hat{\delta} \rangle = \ker \gamma$. The last assertion follows because an invariant subgroup containing $\hat{w}$ contains $A_1 \hat{w} = \hat{u} - \hat{w}$, hence $\hat{u}$ and $\hat{\delta}$.

The exterior-power calculations in (vi) are given in section B. The Smith normal forms there have only unit nonzero elementary divisors, which proves the asserted saturation. The T_0 -invariants also follow directly from $Nw = -u$, $N\delta = \gamma$, and $N\gamma = Nu = 0$. $\square$

2.2. The orbifold curve. Let $B = \mathbb{P}^1$ with affine coordinate t , and put

$$p_1 = 0, \quad p_2 = 1, \quad p_0 = \infty, \quad t_c = t^{-1}, \quad B^\circ = B \setminus \{p_0, p_1, p_2\}.$$

Let $\mathcal{B}$ be the orbifold with underlying curve B , cone orders 3 and 4 at p_1, p_2 , and a cusp at p_0 .

Proposition 2.2 (Uniformization). *There are a copy $\mathfrak{H}_z$ of the upper half-plane, a Fuchsian group $\Delta < \mathrm{PSL}_2(\mathbb{R})$, and a holomorphic quotient map*

$$\pi : \mathfrak{H}_z \longrightarrow B \setminus \{p_0\}$$

with the following properties.

- (i) $\mathfrak{H}_z/\Delta \cong B \setminus \{p_0\}$ as orbifolds.
- (ii) There are $z_j \in \pi^{-1}(p_j)$ and generators g_1, g_2 such that

$$\Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle, \quad g_0 = (g_1 g_2)^{-1}$$

is primitive parabolic, and

$$g_j'(z_j) = e^{-2\pi i/m_j}, \quad (m_1, m_2) = (3, 4).$$

- (iii) *For sufficiently small $r > 0$, the component $U_r \subset \pi^{-1}(0 < |t_c| < r)$ accumulating at the fixed point of g_0 is simply connected, invariant under $\langle g_0 \rangle$, disjoint from all its other Δ -translates, and*

$$U_r/\langle g_0 \rangle \cong \{0 < |t_c| < r\}.$$

Proof. This is the standard uniformization of the hyperbolic triangle orbifold of signature $(0; 3, 4, \infty)$; see [2, Chs. 7, 10]. Choose a hyperbolic triangle with angles $\pi/3, \pi/4, 0$ and take the orientation-preserving subgroup of its reflection group. If the vertices are ordered counter-clockwise, the inverses of the positive rotations about the finite vertices give g_1, g_2 with the stated derivatives, while $(g_1 g_2)^{-1}$ is the primitive cusp generator. The standard disjoint-horodisc theorem for a cofinite Fuchsian group gives (iii). $\square$

The homomorphism $\rho_V : \Delta \rightarrow \mathrm{SL}(V)$ is defined by $\rho_V(g_j) = T_j$. The monodromy on Λ will be the contragredient representation $g_j \mapsto A_j, g_0 \mapsto M_0$.

3. THE PERIOD FAMILY OVER THE PUNCTURED BASE

Let $\rho = e^{\pi i/3}$. We use the classical modular function $j : \mathfrak{H} \rightarrow \mathbb{C}$, normalized by $j(i) = 1728$, and the modular forms $E_4, E_6, \Delta_{\mathrm{mod}}$ of weights 4, 6, 12 [9, Ch. VII].

3.1. The three period functions.

Theorem 3.1 (Periods). *There are holomorphic functions*

$$\tau : \mathfrak{H}_z \rightarrow \mathfrak{H}, \quad \mu, \beta : \mathfrak{H}_z \rightarrow \mathbb{C}$$

satisfying

$$(1) \quad j(\tau) = 1728t \circ \pi,$$

$$(2) \quad \tau \circ g_1 = \frac{\tau - 1}{\tau}, \quad \tau \circ g_2 = -\frac{1}{\tau}, \quad \tau(z_1) = \rho, \quad \tau(z_2) = i,$$

$$(3) \quad \mu \circ g_1 = \frac{1 - \mu}{\tau}, \quad \mu \circ g_2 = 1 + \frac{\mu}{\tau},$$

$$(4) \quad \beta \circ g_1 = \beta + 2 - \frac{6(1 - \mu)^2}{\tau}, \quad \beta \circ g_2 = \beta - 3 - \frac{6\mu^2}{\tau}.$$

These laws imply

$$(5) \quad \tau \circ g_0 = \tau - 1, \quad \mu \circ g_0 = \mu, \quad \beta \circ g_0 = \beta + 1.$$

On the distinguished cusp component U_r , the functions μ and $\beta + \tau$ are bounded and descend holomorphically across $t_c = 0$. The functions τ and μ are uniquely determined by these conditions, whereas the set of functions β satisfying the stated conditions is an affine space under the additive constants.

Proof. The function τ . Let $\mathcal{M}$ denote the modular orbifold, whose coarse coordinate is j and whose orbifold orders at 0, 1728, and ∞ are 3, 2, and ∞ ; the map $j : \mathfrak{H} \rightarrow \mathcal{M}$ is its universal orbifold covering. The coarse map

$$\varphi : B \longrightarrow \mathbb{P}^1, \quad \varphi(t) = 1728t,$$

is a holomorphic orbifold morphism from $(B; 3, 4, \infty)$ to $\mathcal{M}$. We verify the local lifts and their isotropy homomorphisms. Choose source uniformizers s_1, s_2 at p_1, p_2 so that, after multiplying them by invariant holomorphic units,

$$t = s_1^3, \quad t - 1 = s_2^4.$$

Choose target uniformizers ξ_1, ξ_2 at the orbifold points 0, 1728 so that $j = \xi_1^3$ and $j - 1728 = \xi_2^2$, again after multiplication by isotropy-invariant units. The map φ then has local lifts of the form

$$\xi_1 = s_1 \cdot (\text{unit}), \quad \xi_2 = s_2^2 \cdot (\text{unit}).$$

Thus the order-three generator maps to $S_1 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, whose derivative at $\rho = e^{\pi i/3}$ is $e^{-2\pi i/3}$, while the order-four generator maps to $S = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$: the action $s_2 \mapsto -is_2$ induces $\xi_2 \mapsto -\xi_2$. At the cusp, write $q = e^{2\pi i\zeta}$ for the standard modular coordinate, where $\zeta \in \mathfrak{H}$. Since $j = q^{-1}(1 + O(q))$, the equation $j = 1728/t_c$ determines a local lift

$$(6) \quad q = t_c u_1(t_c)$$

with u_1 a holomorphic unit. A clockwise loop about $t_c = 0$ therefore maps to the clockwise modular cusp generator $T^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. Equivalently, g_1, g_2, g_0 map to S_1, S, T^{-1} ; indeed $(S_1 S)^{-1} = T^{-1}$.

The orbifold lifting theorem [2, Chs. 7,10] now gives a holomorphic lift $\tau : \mathfrak{H}_z \rightarrow \mathfrak{H}$ equivariant for the homomorphism $g_1 \mapsto S_1, g_2 \mapsto S$. It satisfies $\tau(z_1) = \rho$. This equivariant lift is unique: two such lifts differ by an element of $\text{PSL}_2(\mathbb{Z})$ centralizing both S_1 and S , and that common centralizer is trivial. The preceding isotropy homomorphisms give (1)–(2). The local lift at p_1 has degree one, so $\tau - \rho$ has order one at z_1 ; the local lift at p_2 has degree two, so $\tau - i$ has order two at z_2 . On U_r , the cusp coordinate in (6) is $q = e^{2\pi i\tau}$.

Choose a branch of $h = (2\pi i)^{-1} \log u_1$ in (6) and set

$$(7) \quad s = \tau - h(t_c \circ \pi).$$

Then $e^{2\pi i s} = t_c \circ \pi$ and $s \circ g_0 = s - 1$. Both $t_c \circ \pi : U_r \rightarrow \{0 < |t_c| < r\}$ and the exponential map from a half-plane are universal coverings with the same deck generator. Hence s is a biholomorphism from U_r onto a half-plane.

The function μ . Define a multiplicative cocycle j on $\Delta \times \mathfrak{H}_z$ by

$$j(g_1, z) = -\tau(z), \quad j(g_2, z) = \tau(z),$$

and the cocycle rule. The identities for the g_1 - and g_2 -orbits of τ show that this is well defined and that $j(g_0, z) = 1$. For an open set $U \subset B$, let $\mathcal{L}(U)$ be the holomorphic functions ν on $\pi^{-1}(U \setminus \{p_0\})$ satisfying

$$\nu \circ g_1 = -\frac{\nu}{\tau}, \quad \nu \circ g_2 = \frac{\nu}{\tau},$$

with the following additional condition when $p_0 \in U$: on the distinguished cusp component, ν descends as a function of t_c and extends holomorphically across $t_c = 0$. These spaces form a sheaf of $\mathcal{O}_B$ -modules, denoted $\mathcal{L}$. We claim

$$(8) \quad \mathcal{L} \cong \mathcal{O}_B(-p_0) \cong \mathcal{O}_{\mathbb{P}^1}(-1).$$

Indeed, since $\mathfrak{H}_z$ is simply connected and $E_6 \circ \tau$ has zeros of even order, choose a holomorphic square root and set

$$F = \left(\frac{E_4^2 E_6^{1/2}}{\Delta_{\text{mod}}} \right) \circ \tau.$$

Choose lifts to $\text{SL}_2(\mathbb{Z})$ of the images of g_1 and g_2 in $\text{PSL}_2(\mathbb{Z})$. The numerator and denominator in the definition of F have total modular weights 11 and 12; hence their quotient has weight -1 . The sign of the chosen square root of E_6 and the sign of an $\text{SL}_2(\mathbb{Z})$ -lift combine into constants $\chi_j \in \{\pm 1\}$ for which

$$F \circ g_j = \frac{\chi_j F}{j(g_j)}.$$

Iterating around $g_1^3 = 1$ gives $\chi_1^3 = 1$, hence $\chi_1 = 1$. At z_2 , the function $E_6^{1/2} \circ \tau$ has a simple zero in the linearizing coordinate s_2 , and $s_2 \circ g_2 = -is_2$. The weight-three automorphy factor at z_2 is $\tau(z_2)^3 = i^3 = -i$, so comparison of the linear terms gives $\chi_2 = 1$. Thus F satisfies the homogeneous laws defining $\mathcal{L}$.

The function F has order 2 at z_1 , order 1 at z_2 , and is t_c^{-1} times a unit at the cusp. If ν is another local homogeneous solution, then ν/F is Δ -invariant and hence a meromorphic function of t . At p_1 a pole of ν/F would lift to a pole of order at most 2 in the coordinate s_1 , hence of order at most $2/3$ in t ; no integral pole order is possible. The same argument at p_2 gives the bound $1/4$. At p_0 , regularity of ν and the simple pole of F force the quotient to vanish. Thus $\mathcal{L}_{p_0} = \mathfrak{m}_{p_0} F$ and F generates elsewhere, proving (8).

The affine transformation laws (3) are consistent with $g_1^3 = g_2^4 = 1$. Explicitly, iteration gives

	z	$g_1 z$	$g_1^2 z$
τ	τ	$(\tau - 1)/\tau$	$-1/(\tau - 1)$
μ	μ	$(1 - \mu)/\tau$	$(\tau - 1 + \mu)/(\tau - 1)$

and

	z	$g_2 z$	$g_2^2 z$	$g_2^3 z$
τ	τ	$-1/\tau$	τ	$-1/\tau$
μ	μ	$1 + \mu/\tau$	$1 - \tau - \mu$	$(1 - \mu)/\tau$

A direct substitution gives

$$\mu \circ (g_1 g_2) = \mu, \quad \text{hence} \quad \mu \circ g_0 = \mu.$$

Consequently the local solutions form a torsor under $\mathcal{L}$. On a disc avoiding the three special points, prescribe a holomorphic function on one lift and extend it by the affine laws. At the cusp the identity $\mu \circ g_0 = \mu$ makes the choice $\mu = 0$ consistent on the distinguished component. On invariant discs at the elliptic points use

$$\mu_1 = \frac{2 - \tau}{3}, \quad \mu_2 = \frac{1 - \tau}{2}.$$

Since $H^1(B, \mathcal{L}) = 0$, the torsor has a global section; since $H^0(B, \mathcal{L}) = 0$, it is unique. At the cusp it is bounded and hence extends holomorphically. Evaluating at the elliptic fixed points gives

$$(9) \quad \mu(z_1) = \frac{1}{1+\rho}, \quad \mu(z_2) = \frac{1-i}{2},$$

and therefore

$$6(1 - \mu(z_1))^2 = 2\tau(z_1), \quad 6\mu(z_2)^2 = -3\tau(z_2).$$

The function β . Put

$$\phi_1 = 2 - \frac{6(1-\mu)^2}{\tau}, \quad \phi_2 = -3 - \frac{6\mu^2}{\tau}.$$

Substitution of the preceding orbit tables gives

$$\begin{aligned} \phi_1 \circ g_1 &= 2 - \frac{6(\tau-1+\mu)^2}{\tau(\tau-1)}, & \phi_1 \circ g_1^2 &= 2 + \frac{6\mu^2}{\tau-1}, \\ \phi_2 \circ g_2 &= -3 + \frac{6(\tau+\mu)^2}{\tau}, & \phi_2 \circ g_2^2 &= -3 - \frac{6(1-\tau-\mu)^2}{\tau}, \\ \phi_2 \circ g_2^3 &= -3 + \frac{6(1-\mu)^2}{\tau}. \end{aligned}$$

A common-denominator calculation yields

$$\sum_{k=0}^2 \phi_1 \circ g_1^k = 0, \quad \sum_{k=0}^3 \phi_2 \circ g_2^k = 0.$$

Moreover (9) gives $\phi_j(z_j) = 0$. Substituting the transformation laws once more gives

$$\beta \circ (g_1 g_2) = \beta - 1, \quad \beta \circ g_0 = \beta + 1.$$

Since $\tau \circ g_0 = \tau - 1$, the function $\beta + \tau$ is g_0 -invariant. Hence the local solutions of (4), with $\beta + \tau$ extending at p_0 , form a torsor under $\mathcal{O}_B$. At the cusp the choice $\beta = -\tau$ is consistent with the induced g_0 -law. On a simply covered disc choose an arbitrary solution, and at p_j use

$$\beta_j = \frac{1}{m_j} \sum_{k=0}^{m_j-1} k \phi_j \circ g_j^k.$$

Since $H^1(B, \mathcal{O}_B) = 0$, a global solution exists, and all solutions differ by constants. $\square$

3.2. Equivariance and the torus family. Define

$$\Pi(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix} = [Z(z) \mid I_2].$$

Regard its rows as the linear forms

$$\sigma_1 = 6\mu\gamma + \tau u + w, \quad \sigma_2 = \beta\gamma + \mu u + \delta \quad \text{in } V \otimes \mathbb{C}.$$

Proposition 3.2 (Equivariance). *For the generators of Δ one has*

$$R_1(z) = \begin{pmatrix} -1/\tau & 0 \\ (1-\mu)/\tau & 1 \end{pmatrix}, \quad R_2(z) = \begin{pmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{pmatrix}, \quad R_0 = I_2,$$

and

$$(10) \quad \Pi(gz) = R_g(z)\Pi(z)M_g, \quad M_g = \rho_V(g)^t.$$

The matrices satisfy $R_{gh}(z) = R_g(hz)R_h(z)$.

Proof. Using the matrices of proposition 2.1,

$$\begin{aligned} T_1\sigma_1 &= (6\mu - 6)\gamma + (1 - \tau)u - \tau w, \\ T_1\sigma_2 &= (\beta + 2)\gamma + (1 - \mu)u + (1 - \mu)w + \delta, \\ T_2\sigma_1 &= (6\mu + 6\tau)\gamma - u + \tau w, \\ T_2\sigma_2 &= (\beta + 6\mu - 3)\gamma + u + \mu w + \delta. \end{aligned}$$

Substituting (2)–(4) gives

$$\sigma_1(g_1z) = -\tau^{-1}T_1\sigma_1, \quad \sigma_2(g_1z) = T_1\sigma_2 + (1 - \mu)\tau^{-1}T_1\sigma_1,$$

and the analogous formulas for g_2 . For g_0 one obtains $\sigma_i(g_0z) = T_0\sigma_i(z)$. Evaluating on Λ gives (10); the cocycle rule follows by uniqueness. $\square$

Corollary 3.3 (Choice of the additive constant). *The function β may be chosen so that*

$$(11) \quad D := \Im\beta - \frac{6(\Im\mu)^2}{\Im\tau} < 0 \quad \text{on } \mathfrak{H}_z.$$

For this choice, $L(z) = \Pi(z)\Lambda$ is a lattice in $\mathbb{C}^2$ for every $z \in \mathfrak{H}_z$.

Proof. As a real linear map $\Lambda \otimes \mathbb{R} \rightarrow \mathbb{C}^2$,

$$(12) \quad \det_{\mathbb{R}} \Pi(z) = -\det \Im Z(z) = \Im\tau(z) D(z).$$

For both generators,

$$|\det_{\mathbb{C}} R_g(z)|^2 = |\tau(z)|^{-2}, \quad \Im\tau(gz) = \Im\tau(z)|\tau(z)|^{-2}.$$

Taking real determinants in (10) and using $\det M_g = 1$, comparison with (12) therefore gives $D(gz) = D(z)$. Thus D is Δ -invariant. Since τ, μ, β are holomorphic at the elliptic fixed points, it descends continuously to $B \setminus \{p_0\}$. At the cusp, $\beta + \tau$ and μ are bounded while $\Im\tau \rightarrow \infty$, so $D \rightarrow -\infty$. Hence D is bounded above on $\mathfrak{H}_z$ modulo Δ . Replacing β by $\beta + c_0$ adds $\Im c_0$ to D ; a sufficiently negative imaginary constant gives (11). Formula (12) then shows that $\Pi(z)$ is a real isomorphism for every z , so its integral columns form a lattice in $\mathbb{C}^2$. $\square$

Define

$$\mathcal{T} = (\mathfrak{H}_z \times \mathbb{C}^2)/\Lambda, \quad t_\lambda(z, \zeta) = (z, \zeta + \Pi(z)\lambda).$$

The lattice action is free and properly discontinuous. Indeed, over a compact subset of $\mathfrak{H}_z$, the real isomorphisms $\Pi(z)$ and their inverses are uniformly bounded. Hence, for compact subsets of $\mathfrak{H}_z \times \mathbb{C}^2$, only finitely many lattice translations can produce an intersection. Thus $\mathcal{T}$ is a complex manifold and the projection $\mathcal{T} \rightarrow \mathfrak{H}_z$ is a holomorphic family of compact complex two-tori. We use the same notation t_λ for the corresponding fibre loop in fundamental-group calculations. The maps

$$\tilde{g}(z, \zeta) = (gz, R_g(z)\zeta)$$

normalize the lattice action. Let $\mathfrak{H}_z^\circ$ be the complement of the Δ -orbits of z_1, z_2 .

Corollary 3.4. *The quotient*

$$J = (\mathcal{T}|_{\mathfrak{H}_z^\circ})/\Delta \longrightarrow B^\circ$$

is a proper holomorphic submersion with fibres complex two-tori and a holomorphic zero section. In the marking $H_1(F; \mathbb{Z}) = \Lambda$, the clockwise meridians about p_1, p_2, p_0 have monodromies A_1, A_2, M_0 .

Proof. The action on $\mathfrak{H}_z^\circ$ is free and properly discontinuous, hence so is the induced action on the torus family. Over a simply connected disc $U \Subset B^\circ$, choose one lift $\tilde{U} \subset \mathfrak{H}_z^\circ$. The quotient above U is then the family $(\tilde{U} \times \mathbb{C}^2)/\Lambda$, with lattice generators depending holomorphically on the base. A fixed real fundamental parallelepiped for $\Lambda \otimes \mathbb{R}/\Lambda$, transported by $\Pi(z)$, gives a compact set surjecting onto the inverse image of every compact subset of U . Thus the map is proper locally on the base, hence proper globally. The same local description shows that it is a holomorphic submersion and that the zero section descends.

Parallel transport in the trivial marked family over $\mathfrak{H}_z^\circ$, followed by the identification through $\tilde{g}$, acts on the lattice by M_g^{-1} , which is A_j for g_j and M_0 for g_0 . $\square$

3.3. The period normal form at the cusp. On U_r put $b = \beta + \tau$. The functions μ, b, h are holomorphic in t_c , and with

$$C(t_c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b(t_c) - h(t_c) & \mu(t_c) \end{pmatrix}$$

one has

$$(13) \quad Z(z) = s(z)B_0 + C(t_c \circ \pi(z)).$$

Consequently, for $\bar{\lambda} \in \bar{\Lambda} \cong \mathbb{Z}^2$,

$$(14) \quad \exp(2\pi i Z \bar{\lambda}) = t_c^{B_0 \bar{\lambda}} c_{\bar{\lambda}}(t_c), \quad c_{\bar{\lambda}}(t_c) = \exp(2\pi i C(t_c) \bar{\lambda}).$$

4. THE TORIC FILLING AT THE CUSP

4.1. The infinite toric threefold. Let

$$N' = \mathbb{Z}^2 \oplus \mathbb{Z}, \quad M' = \text{Hom}(N', \mathbb{Z}),$$

and write points of $N'_\mathbb{R}$ as (y, y_3) . Let $\mathcal{T}_{A_2}$ be the triangulation of $\mathbb{R}^2$ with vertices $\mathbb{Z}^2$ and triangles

$$\{v, v + e_1, v + e_2\}, \quad \{v + e_1, v + e_2, v + e_1 + e_2\}.$$

For every cell P let $\hat{P}$ be the cone over $P \times \{1\}$, and let Σ be the fan consisting of these cones and their faces.

Lemma 4.1 (The toric model). *The collection Σ is a smooth countable fan with support $(\mathbb{R}^2 \times \mathbb{R}_{>0}) \cup \{0\}$. The associated toric space $Y = Y_\Sigma$ is a connected Hausdorff second-countable complex threefold. On its dense torus $(\mathbb{C}^*)^3$, use coordinates (x_1, x_2, t_c) , where t_c is the character corresponding to $(0, 0, 1) \in M'$. If a maximal cone has vertices v_0, v_1, v_2 and m_0, m_1, m_2 is the dual basis, then*

$$U_\sigma = \text{Spec } \mathbb{C}[\hat{\sigma}^\vee \cap M'] \cong \mathbb{C}^3, \quad z_i = \chi^{m_i},$$

and

$$(15) \quad t_c = z_0 z_1 z_2.$$

The divisor $Y_0 = t_c^{-1}(0)$ is reduced and equals $\bigcup_{v \in \mathbb{Z}^2} E_v$. Each E_v is the toric del Pezzo surface dP_6 , and its toric boundary is the hexagon of six (-1) -curves indexed by the six edges through v .

Proof. For the lower triangle the determinant of the three primitive generators $(v, 1), (v + e_1, 1), (v + e_2, 1)$ is 1; for the upper triangle it is -1 . Thus every cone is unimodular. Two cones meet in the cone over the common face of their cells, and the support is as stated because a point (y, y_3) with $y_3 > 0$ lies over the cell containing y/y_3 . Glue the affine toric charts U_σ along the open subspaces $U_{\sigma \cap \tau}$. The fan condition implies that if two points represented in U_σ and U_τ are identified through a chain of charts, then they are already identified in $U_{\sigma \cap \tau}$. Consequently the union of any two charts is the toric space of the finite fan consisting of the faces of σ and τ ; it is Hausdorff by the standard separation argument for fans [5, Secs. 1.4, 3.1]. Thus the full gluing is Hausdorff. Every chart is smooth, the atlas is countable, and all charts meet the dense torus. Hence Y is a connected second-countable complex threefold.

The height character $(0, 0, 1)$ is nonnegative on the fan and takes value one on every ray $(v, 1)$. Hence its divisor is $\sum_v E_v$ with multiplicity one. In a maximal chart, $(0, 0, 1) = m_0 + m_1 + m_2$, proving (15) and reducedness. The star of the ray $(v, 1)$, modulo that ray, is the complete smooth fan with rays in the six directions $\pm e_1, \pm e_2, \pm(e_1 - e_2)$, which is the fan of dP_6 . $\square$

4.2. The lattice action. Identify the cocharacter lattice of $(x_1, x_2, 1)$ with Λ_{tor} by $e_1 \leftrightarrow \widehat{w}$ and $e_2 \leftrightarrow \widehat{\delta}$. For $\bar{\lambda} \in \bar{\Lambda}$ define

$$\varphi_{\bar{\lambda}}(y, y_3) = (y + y_3 B_0 \bar{\lambda}, y_3).$$

It preserves the fan and induces a toric automorphism $\Phi_{\bar{\lambda}}$ of Y . On the dense torus,

$$\Phi_{\bar{\lambda}}(x, t_c) = (t_c^{B_0 \bar{\lambda}} x, t_c).$$

Using (14), set

$$(16) \quad \Psi_{\bar{\lambda}}(x, t_c) = (c_{\bar{\lambda}}(t_c) t_c^{B_0 \bar{\lambda}} x, t_c).$$

The formula extends holomorphically to Y because it is the product of $\Phi_{\bar{\lambda}}$ with a holomorphic torus translation. The shears $\varphi_{\bar{\lambda}}$ fix the horizontal cocharacter lattice $\mathbb{Z}^2 \oplus 0$, so the toric automorphisms commute with these translations. Since $C(t_c)(\bar{\lambda} + \bar{\mu}) = C(t_c)\bar{\lambda} + C(t_c)\bar{\mu}$, one has $\Psi_{\bar{\lambda}}\Psi_{\bar{\mu}} = \Psi_{\bar{\lambda}+\bar{\mu}}$. Thus (16) defines an action of $\bar{\Lambda} \cong \mathbb{Z}^2$. The compact horizontal torus $T_f = \Lambda_{\text{tor}} \otimes S^1$ commutes with this action and therefore acts on the quotient constructed below, preserving t_c .

For a torus point with $0 < |t_c| < 1$ define

$$y(p) = \frac{\log |x(p)|}{\log |t_c(p)|} \in \mathbb{R}^2.$$

Let θ_i^σ be the barycentric coordinates of a maximal triangle σ , and put

$$\sigma(\eta) = \{y : \theta_i^\sigma(y) > -\eta, \ i = 0, 1, 2\}.$$

For $S \geq 1$ write $U_\sigma(S) = \{p \in U_\sigma : |z_i(p)| < S\}$.

Lemma 4.2 (Toric chart estimates). *For a torus point p ,*

$$p \in U_\sigma(S) \iff y(p) \in \sigma \left(\frac{\log S}{|\log |t_c(p)||} \right).$$

Every point of $|t_c| < 1$ lies in the closed unit polydisc of some maximal toric chart. In particular the sets $U_\sigma(2)$ cover Y_ε , and every compact subset has a finite subcover.

Proof. In the coordinates of lemma 4.1,

$$\log |z_i(p)| = \log |t_c(p)| \theta_i^\sigma(y(p)),$$

which gives the first assertion because $\log |t_c| < 0$. For a torus point, choose a triangle containing $y(p)$.

Let now p lie in the orbit associated with the cone over a cell P . For $\bar{m} \in M' \cap \widehat{P}^\perp$, the assignment $\bar{m} \mapsto \log |\chi^{\bar{m}}(p)|$ is a real linear functional; under the natural pairing it is represented by a vector $\text{Log}_P(p)$ in $N'_\mathbb{R} / \text{span}_\mathbb{R} \widehat{P}$. For a maximal triangle $\sigma \supset P$, let $\bar{\sigma}$ be the image of its cone in this quotient. The cones $\bar{\sigma}$, as σ ranges over the maximal triangles containing P , cover the quotient vector space: if v is a quotient vector and q_0 lies in the relative interior of $P \times \{1\}$, then for sufficiently small $u > 0$ the height-one normalization of $q_0 + u\tilde{v}$ lies in a triangle containing P . Choose σ with $-\text{Log}_P(p) \in \bar{\sigma}$. If $\bar{m}$ is a dual coordinate on the orbit, then $\log |\chi^{\bar{m}}(p)| = \langle \bar{m}, \text{Log}_P(p) \rangle$; the chosen inclusion therefore makes every nonzero chart coordinate have modulus at most one. This proves the second assertion. The last assertion follows from compactness. $\square$

Theorem 4.3 (Cusp filling). *For sufficiently small $\varepsilon > 0$, the action of $\bar{\Lambda}$ on*

$$Y_\varepsilon = \{|t_c| < \varepsilon\} \subset Y$$

is free and properly discontinuous. The quotient

$$N_0 = Y_\varepsilon / \bar{\Lambda}$$

is a connected Hausdorff complex threefold, and t_c descends to a proper surjective holomorphic map

$$f_0 : N_0 \longrightarrow \Delta_0 = \{|t_c| < \varepsilon\}.$$

It is a submersion away from 0, and its nonzero fibres are the tori of corollary 3.4. The central fibre

$$W = f_0^{-1}(0)$$

is reduced, irreducible and has normal crossings. Its normalization is dP_6 ; the normalization map identifies opposite sides of the anticanonical hexagon. The double locus consists of three rational curves through two triple points.

Moreover

$$\pi_1(N_0) \cong \bar{\Lambda},$$

the inclusion of a nearby fibre sends Λ to $\bar{\Lambda} = \Lambda/\Lambda_{\text{tor}}$, and the loop obtained by holding $(x_1, x_2) = (1, 1)$ and circling $t_c = 0$ is null-homotopic in N_0 .

Proof. Put

$$R(t_c) = -2\pi\Im C(t_c), \quad B_{t_c} = B_0 + \frac{R(t_c)}{\log |t_c|}.$$

Choose ε so small that $\|B_{t_c} - B_0\| \leq \frac{1}{2}$ for $0 < |t_c| < \varepsilon$. Since B_0 is orthogonal in the chosen bases,

$$(17) \quad |B_{t_c}\bar{\lambda}| \geq \frac{1}{2}|\bar{\lambda}|.$$

Freeness. On the dense torus, a fixed point of $\Psi_{\bar{\lambda}}$ with $\bar{\lambda} \neq 0$ would satisfy

$$\log |t_c| B_0\bar{\lambda} + R(t_c)\bar{\lambda} = 0,$$

which contradicts the choice of ε . On Y_0 , the action sends the orbit attached to a cell P to the orbit attached to $P + B_0\bar{\lambda}$. No nonzero translation preserves a bounded cell, so there are no fixed points there either.

Proper discontinuity. Equation (16) gives, on the dense torus,

$$(18) \quad y(\Psi_{\bar{\lambda}}p) = y(p) + B_{t_c(p)}\bar{\lambda}.$$

Fix maximal triangles σ, σ' . By lemma 4.2, the positions of torus points in $U_\sigma(2) \cap Y_\varepsilon$ and $U_{\sigma'}(2) \cap Y_\varepsilon$ lie in bounded sets $\sigma(\eta_0)$ and $\sigma'(\eta_0)$, where $\eta_0 = \log 2/|\log \varepsilon|$. If $\Psi_{\bar{\lambda}}U_\sigma(2)$ meets $U_{\sigma'}(2)$, density of the torus in every chart supplies a torus point in the intersection. Then

$$\frac{1}{2}|\bar{\lambda}| \leq |B_{t_c}\bar{\lambda}| \leq \sup\{|a - b| : a \in \sigma'(\eta_0), b \in \sigma(\eta_0)\},$$

so only finitely many $\bar{\lambda}$ are possible. A compact subset is covered by finitely many $U_\sigma(2)$; therefore only finitely many of its translates meet it. This is proper discontinuity.

The orbit space is Hausdorff and the orbit map is a covering. If K is a compact neighbourhood of a point p , proper discontinuity leaves only finitely many nontrivial translates of K that meet K ; freeness and Hausdorffness allow K to be shrunk to an open neighbourhood disjoint from all its nontrivial translates. The same finite-translate argument separates two distinct orbits. Since all deck transformations are biholomorphic, the quotient is a complex manifold. The set Y_ε is connected: in the coordinates of a maximal chart, $U_\sigma \cap Y_\varepsilon = \{|z_0 z_1 z_2| < \varepsilon\}$ is star-shaped and meets the connected dense set $(\mathbb{C}^*)^2 \times \{0 < |t_c| < \varepsilon\}$. It is second countable, and the open quotient map shows that N_0 is second countable as well. Hence N_0 is a connected complex threefold.

Properness. Fix $0 < \varepsilon_1 < \varepsilon$. The matrices B_{t_c} and their inverses are uniformly bounded for $0 < |t_c| \leq \varepsilon_1$. Given a torus point p , put $u = B_{t_c}^{-1}y(p)$ and choose $\bar{\lambda} \in \mathbb{Z}^2$ with $|u + \bar{\lambda}|_\infty \leq 1/2$. Then

$$|y(\Psi_{\bar{\lambda}}p)| = |B_{t_c}(u + \bar{\lambda})| \leq C$$

for a constant independent of p . Choose the finite set of maximal triangles meeting the closed ball $\bar{B}(0, C+2)$ and, in each corresponding chart, the compact unit polydisc $\{|z_i| \leq 1, |t_c| \leq \varepsilon_1\}$. By lemma 4.2, every translated torus point above the closed disc lies in their union.

For a point of Y_0 in the orbit attached to a cell P , choose a vertex v of P . Unimodularity of B_0 gives $\bar{\lambda}$ with $B_0\bar{\lambda} = -v$, so the translated cell contains 0 and is a face of one of the finitely

many triangles adjacent to 0. Enlarging the preceding compact union by the corresponding closed chart polydiscs gives a compact set K_0 meeting every orbit over $|t_c| \leq \varepsilon_1$. Therefore $f_0^{-1}(|t_c| \leq \varepsilon_1)$ is the compact image of K_0 . Any compact subset of the open disc lies in such a closed subdisc, proving that f_0 is proper.

The fibres and local structure. For $t_c \neq 0$, choose s with $e^{2\pi i s} = t_c$. The exponential map $\mathbb{C}^2 \rightarrow (\mathbb{C}^*)^2$ identifies the quotient fibre with

$$\mathbb{C}^2 / (\mathbb{Z}^2 + (sB_0 + C(t_c))\mathbb{Z}^2) = \mathbb{C}^2 / \Pi(z)\Lambda,$$

where $s = s(z)$; this is the torus family of corollary 3.4. These fibres exist for every $0 < |t_c| < \varepsilon$, while Y_0 is nonempty, so f_0 is surjective. On $t_c \neq 0$ the character $t_c : (\mathbb{C}^*)^3 \rightarrow \mathbb{C}^*$ is a submersion, and the quotient map is locally biholomorphic. Hence f_0 is a submersion away from the origin. Finally, suppose that exactly the coordinates indexed by a nonempty set $I \subset \{0, 1, 2\}$ vanish at a chosen point. The product of the remaining coordinates is a holomorphic unit; absorbing that unit into one coordinate indexed by I , (15) gives local coordinates on N_0 in which f_0 is $z_0, z_0 z_1$, or $z_0 z_1 z_2$, according as $|I| = 1, 2, 3$. Thus W is reduced and has normal crossings.

Because $B_0 \bar{\Lambda} = \mathbb{Z}^2$, the action on the height-one lattice is the full translation action. Every cell has a vertex, so every toric orbit in Y_0 is equivalent to an orbit contained in E_0 ; hence the map $E_0 \rightarrow W$ is surjective. It is injective on the open torus orbit. Two orbits in E_0 can be identified only when cells P, P' containing 0 satisfy $P' = P + v$ for a nonzero integral translation. For edges this forces $P = \{0, v\}$ and $P' = \{0, -v\}$. No nonzero translation preserves an edge, so each boundary curve maps isomorphically to its image. Thus the six boundary curves are identified in the three opposite pairs and no other curve identifications occur. For triangles, the six triangles through 0 fall into two translation orbits, alternating around the hexagon; their images are the two triple points. The fibres of $E_0 \rightarrow W$ are therefore finite. The map is proper because E_0 is compact and W is Hausdorff; a proper holomorphic map with finite fibres is finite. Since E_0 is irreducible and maps onto W , the surface W is irreducible. The finite map $E_0 \rightarrow W$ is generically one-to-one, with normal source and reduced target; it is therefore the normalization. This proves the stated description of the double and triple loci.

Fundamental group. The toric threefold Y is simply connected. For $n \geq 1$, let Σ_n be the finite subfan consisting of the cones over the cells of $\mathcal{T}_{A_2}$ contained in the square $[-n, n]^2$, together with all their faces. The corresponding toric spaces Y_{Σ_n} form an increasing open cover of Y . Each contains the rays through $(0, 0, 1), (1, 0, 1), (0, 1, 1)$, which generate N' over $\mathbb{Z}$. The standard toric fundamental-group formula [5, Sec. 3.2] therefore gives $\pi_1(Y_{\Sigma_n}) = 0$. Every loop has compact image and hence lies in some Y_{Σ_n} , so $\pi_1(Y) = 0$.

For $0 < u \leq 1$, the torus element $(1, 1, u) \in (\mathbb{C}^*)^3$ acts on Y and satisfies $t_c((1, 1, u) \cdot p) = u t_c(p)$. A loop in Y_ε bounds a disc in Y ; for $u > 0$ sufficiently small, translating that disc by $(1, 1, u)$ places it in Y_ε . On the boundary, the path of torus elements $(1, 1, u_s)$, with u_s decreasing from 1 to u , gives a homotopy inside Y_ε from the original loop to the translated one. Hence Y_ε is simply connected. Therefore the covering $Y_\varepsilon \rightarrow N_0$ is universal and $\pi_1(N_0) = \bar{\Lambda}$.

On a nonzero fibre the covering $(\mathbb{C}^*)^2 \rightarrow f_0^{-1}(t_c)$ has deck group $\bar{\Lambda}$, so the induced map on fundamental groups is the projection $\Lambda \rightarrow \bar{\Lambda}$. Finally, the curve $\{(1, 1, t_c) : |t_c| < \varepsilon\}$ is a holomorphic disc in a toric chart; its boundary is the stated meridian, which is therefore null-homotopic. $\square$

Proposition 4.4 (Identification over the punctured cusp disc). *After decreasing ε if necessary, put $U_\varepsilon = U_r \cap \pi^{-1}(D_0^*)$. The map*

$$(19) \quad E_0^{\text{exp}} : U_\varepsilon \times \mathbb{C}^2 \longrightarrow Y_\varepsilon \cap (\mathbb{C}^*)^3, \quad E_0^{\text{exp}}(z, \zeta) = (e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}, e^{2\pi i s(z)}),$$

induces a biholomorphism

$$(20) \quad G_0 : J|_{D_0^*} \xrightarrow{\sim} N_0 \setminus W, \quad D_0^* = \{0 < |t_c| < \varepsilon\}.$$

It identifies the marking $H_1(F; \mathbb{Z}) = \Lambda$ on both sides and carries the zero section to the section represented by $(x_1, x_2) = (1, 1)$. The latter extends across $t_c = 0$ as a holomorphic section of f_0 , meeting W in its smooth locus.

Proof. Let $\tilde{g}_0(z, \zeta) = (g_0 z, \zeta)$; the latter formula follows from $R_0 = I_2$. Let H be the subgroup generated by $\tilde{g}_0$ and the translations t_λ with $\lambda \in \Lambda_{\text{tor}}$. It is normal in $\Lambda \rtimes \langle g_0 \rangle$: conjugation by $\tilde{g}_0$ preserves the translations by Λ_{tor} , while

$$t_\mu \tilde{g}_0 t_\mu^{-1} = t_{\mu - M_0 \mu} \tilde{g}_0 \in H \quad (\mu \in \Lambda),$$

because $(M_0 - I)\Lambda \subset \Lambda_{\text{tor}}$.

The map $s : U_\varepsilon \rightarrow \{\Im s > c\}$ is a biholomorphism and satisfies $s \circ g_0 = s - 1$. Hence the last coordinate in (19) is the universal covering of the punctured disc, with deck group generated by g_0 . The first two coordinates are the ordinary exponential covering $\mathbb{C}^2 \rightarrow (\mathbb{C}^*)^2$, whose deck lattice is $\mathbb{Z}^2 = \Pi(z)\Lambda_{\text{tor}}$. It follows that E_0^{exp} is a surjective local biholomorphism and that its fibres are exactly the H -orbits.

Write $\lambda = \lambda_{\text{tor}} + \lambda'$ and let $\bar{\lambda}$ be its image in $\bar{\Lambda}$. By (13) and (14),

$$E_0^{\text{exp}}(z, \zeta + \Pi(z)\lambda) = \Psi_{\bar{\lambda}}(E_0^{\text{exp}}(z, \zeta)).$$

Thus E_0^{exp} identifies $(U_\varepsilon \times \mathbb{C}^2)/H$ with $Y_\varepsilon \cap (\mathbb{C}^*)^3$ and intertwines the residual action $(\Lambda \rtimes \langle g_0 \rangle)/H \cong \bar{\Lambda}$ with the action Ψ of (16). Since U_ε is the only component of $\pi^{-1}(D_0^*)$ stabilized by g_0 and its other Δ -translates are disjoint, the quotient by the full residual action is precisely (20).

The construction sends the cycles in Λ_{tor} to the fundamental cycles of $(\mathbb{C}^*)^2$ and the quotient $\Lambda/\Lambda_{\text{tor}}$ to the deck group of a fibre of $N_0 \setminus W$; hence it preserves the marking. Finally $E_0^{\text{exp}}(z, 0) = (1, 1, t_c)$. In the toric chart corresponding to the triangle with vertices $0, e_1, e_2$, the coordinates are $(t_c/(x_1 x_2), x_1, x_2)$, so

$$\{(1, 1, t_c) : |t_c| < \varepsilon\}$$

is a holomorphic disc meeting the divisor $t_c = 0$ transversely at a point of the open orbit of E_0 . Since every $\Psi_{\bar{\lambda}}$ preserves t_c and the disc contains exactly one point over each value of t_c , the quotient map is injective on this disc. Its image is therefore the required holomorphic section. $\square$

Proposition 4.5 (Boundary of the cusp piece). *For $0 < \rho < \varepsilon$, the boundary $\mathcal{M}_0 = f_0^{-1}(|t_c| = \rho)$ is the mapping torus of the monodromy M_0 on F . With a_0 the toric meridian,*

$$\pi_1(\mathcal{M}_0) = \Lambda \rtimes_{M_0} \mathbb{Z}\langle a_0 \rangle.$$

The inclusion $\mathcal{M}_0 \hookrightarrow N_0$ sends $\lambda \mapsto \bar{\lambda} \in \bar{\Lambda}$ and $a_0 \mapsto 1$; its kernel is the normal closure of $\Lambda_{\text{tor}} \cup \{a_0\}$.

Proof. Ehresmann's theorem identifies the restriction over the circle with the mapping torus of the geometric monodromy, whose translation part is isotopic to zero and whose linear part is M_0 . The map on the fibre fundamental group is the projection established in theorem 4.3, and the meridian is null-homotopic there. The resulting quotient of the displayed semidirect product is $\Lambda/\Lambda_{\text{tor}} = \bar{\Lambda}$, which is $\pi_1(N_0)$; hence the stated normal subgroup is exactly the kernel. $\square$

5. THE FINITE FILLINGS AND THE COMPACT COMPLEX THREEFOLD

5.1. Free logarithmic transforms. For $j = 1, 2$, choose a g_j -invariant disc $\Delta_j \ni z_j$ and a coordinate s_j such that

$$s_j(z_j) = 0, \quad s_j \circ g_j = e^{-2\pi i/m_j} s_j.$$

The quotient coordinate t_j on $D_j = \Delta_j/\langle g_j \rangle$ is defined by $t_j \circ \pi = s_j^{m_j}$. Let

$$J'_j = (\Delta_j \times \mathbb{C}^2)/\Lambda$$

be the restriction of the marked torus family. Define

$$(21) \quad \tilde{g}_j^{\log}(z, \zeta) = \left(g_j z, R_j(z)\zeta + \frac{1}{m_j} \Pi(g_j z) v_j \right).$$

The identity $R_j(z)\Pi(z) = \Pi(g_j z)A_j$ implies

$$\tilde{g}_j^{\log} t_\lambda = t_{A_j \lambda} \tilde{g}_j^{\log},$$

so (21) descends to J'_j . In the smooth trivialization

$$\Delta_j \times (\Lambda \otimes \mathbb{R})/\Lambda \longrightarrow J'_j, \quad (z, x) \longmapsto [(z, \Pi(z)x)],$$

it is

$$(22) \quad (z, x) \longmapsto (g_j z, A_j x + v_j/m_j).$$

Lemma 5.1 (Fixed points of an affine torus map). *Let $A \in \mathrm{GL}(\Lambda)$ have finite order and let $b \in \Lambda \otimes \mathbb{Q}$. The affine map $x \mapsto Ax + b$ of $(\Lambda \otimes \mathbb{R})/\Lambda$ has a fixed point if and only if*

$$\phi(b) \in \mathbb{Z} \quad \text{for every } \phi \in V \text{ satisfying } \phi \circ A = \phi.$$

Proof. A fixed point exists exactly when $b \in (A - I)(\Lambda \otimes \mathbb{R}) + \Lambda$. Necessity is immediate. For the converse, average the powers of A to obtain the rational projection p onto $\ker(A - I)_{\mathbb{R}}$ with kernel $\mathrm{im}(A - I)_{\mathbb{R}}$. The lattice $p(\Lambda)$ is dual to the invariant lattice $\{\phi \in V : \phi A = \phi\}$. The integrality assumption therefore gives $pb = p\lambda$ for some $\lambda \in \Lambda$, so $b - \lambda \in \mathrm{im}(A - I)_{\mathbb{R}}$. $\square$

Theorem 5.2 (The two logarithmic fillings). *The map (21) generates a free cyclic action of order m_j on J'_j . Its quotient*

$$N_j = J'_j / \langle \tilde{g}_j^{\log} \rangle$$

is a smooth complex threefold, and $s_j^{m_j}$ descends to a proper holomorphic map

$$f_j : N_j \longrightarrow D_j$$

with divisor

$$f_j^*(p_j) = m_j S_j,$$

where S_j is a smooth compact surface.

Over the punctured disc there is a biholomorphism

$$G_j : N_j \setminus S_j \longrightarrow J|_{D_j^*}.$$

Under this biholomorphism the zero section of J is represented on the cover by

$$(23) \quad \varsigma_j(z) = -\frac{\log s_j(z)}{2\pi i} \Pi(z) v_j.$$

Following this section around s_j once counterclockwise changes its lift by $-\Pi(z)v_j$.

Proof. Because $A_j v_j = v_j$, the m_j -th power of (22) is translation by $v_j \in \Lambda$, hence the identity. A nontrivial power can have a fixed point only over z_j . A direct calculation gives

$$V^{T_1} = V^{T_1^2} = \langle \gamma, \psi_1 \rangle, \quad \psi_1 = 2u + w + 3\delta,$$

and

$$V^{T_2} = V^{T_2^2} = V^{T_2^3} = \langle \gamma, \psi_2 \rangle, \quad \psi_2 = u + w + 2\delta.$$

The k -th power of (22) has translation part kv_j/m_j . For $v_1 = \varepsilon$ one has $\gamma(v_1) = 1$ and $\psi_1(v_1) = 0$; hence for $k = 1, 2$ the value $k\gamma(v_1)/3$ is not integral. For $v_2 = -\varepsilon'$ one has $\gamma(v_2) = -1$ and $\psi_2(v_2) = 0$; hence for $k = 1, 2, 3$ the value $-k/4$ is not integral. The fixed-point criterion in lemma 5.1 therefore excludes fixed points for every nontrivial power.

A free finite holomorphic action has a smooth quotient. The function $s_j^{m_j}$ is invariant, properness follows from properness of $J'_j \rightarrow \Delta_j$ and finiteness of the quotient, and its divisor is m_j times the reduced central fibre.

On Δ_j^* define the section

$$\sigma_j(z) = \frac{\log s_j(z)}{2\pi i} \Pi(z) v_j.$$

Changing the logarithm changes σ_j by a period, so it is well defined on the torus family. Translation by σ_j conjugates $\tilde{g}_j^{\log}$ to the linear deck transformation $\tilde{g}_j$. Indeed $s_j(g_j z) = e^{-2\pi i/m_j} s_j(z)$ and $R_j(z) \Pi(z) v_j = \Pi(g_j z) A_j v_j = \Pi(g_j z) v_j$, so the translation terms cancel. This gives G_j . The zero section in N_j -coordinates is $-\sigma_j$, proving (23) and the stated drift. $\square$

Proposition 5.3 (Topology of the logarithmic pieces). *The radial contraction of the s_j -disc is equivariant and induces a deformation retraction $N_j \simeq S_j$. Its fundamental group has the presentation*

$$\pi_1(N_j) = \langle \Lambda, \widehat{g}_j \mid \widehat{g}_j \lambda \widehat{g}_j^{-1} = A_j \lambda, \widehat{g}_j^{m_j} = t_{v_j} \rangle.$$

For the boundary mapping torus $M_j = f_j^{-1}(|t_j| = \rho)$ one has $\pi_1(M_j) = \Lambda \rtimes_{A_j} \mathbb{Z}\langle \widehat{g}_j \rangle$, and the kernel of $\pi_1(M_j) \rightarrow \pi_1(N_j)$ is normally generated by $\widehat{g}_j^{m_j} t_{-v_j}$.

Proof. Before quotienting, the local model is smoothly $A_0^{(j)} \times \Delta_{s_j}$, where $A_0^{(j)} = \mathbb{C}^2/L(z_j)$, and the generator acts by

$$(x, s_j) \mapsto (A_j x + v_j/m_j, e^{-2\pi i/m_j} s_j).$$

Radial contraction of the disc is equivariant. The universal cover of S_j is $\Lambda \otimes \mathbb{R}$, and the group generated by translations t_λ and the affine lift $x \mapsto A_j x + v_j/m_j$ has the displayed relations; its m_j -th power is t_{v_j} . On the boundary, the lift of one m_j -th of the clockwise base circle is an infinite-order element $\widehat{g}_j$. Traversing it m_j times gives the lattice translation v_j together with the full boundary circle of the covering disc. Killing that boundary circle on filling the disc gives exactly the normal relation $\widehat{g}_j^{m_j} t_{-v_j}$. $\square$

5.2. Gluing. Shrink D_0, D_1, D_2 so that they are pairwise disjoint and

$$B = B^\circ \cup D_0 \cup D_1 \cup D_2.$$

Use the biholomorphism G_0 of proposition 4.4 and the biholomorphisms G_1, G_2 of theorem 5.2. Define

$$X = (J \sqcup N_0 \sqcup N_1 \sqcup N_2) / \sim,$$

where points in $J|_{D_0^*}$ are identified through G_0 , and points in $N_j \setminus S_j$ through G_j .

Theorem 5.4 (The compact complex threefold). *The space X is a connected compact Hausdorff complex threefold. The local maps glue to a proper surjective holomorphic map*

$$f : X \longrightarrow B = \mathbb{P}^1$$

with connected fibres. Over B° its fibres are complex two-tori; $f^{-1}(p_0) = W$ is the reduced normal crossings surface of theorem 4.3; and

$$f^*(p_1) = 3S_1, \quad f^*(p_2) = 4S_2.$$

Proof. The three overlap regions in J are disjoint. Hence every nonsingleton equivalence class consists of exactly two points, and the quotient maps from the four pieces are open embeddings. Their transition maps are the biholomorphisms $G_0^{\pm 1}, G_1^{\pm 1}, G_2^{\pm 1}$, so they define a complex atlas on X . Since the four pieces are second countable and their images are open, X is second countable.

Points with different images in B are separated by inverse images of disjoint base neighbourhoods. Points over the same base point lie in one of the Hausdorff local models, proving that X is Hausdorff. Properness is local on the base and holds for each of the four local maps; therefore f is proper and $X = f^{-1}(B)$ is compact. Connectedness and the fibre descriptions follow from the corresponding properties of the local models. $\square$

6. THE FUNDAMENTAL GROUP

Orient every meridian clockwise. Let ρ_j be the lift of the meridian about p_j along the zero section of J . Since B° is a pair of pants,

$$\pi_1(J) = \Lambda \rtimes F(\rho_1, \rho_2), \quad \rho_j \lambda \rho_j^{-1} = A_j \lambda, \quad \rho_0 = (\rho_1 \rho_2)^{-1}.$$

Lemma 6.1 (Relations imposed by the fillings). *Under the gluing maps, the clockwise meridian lifts satisfy*

$$(24) \quad \rho_1^3 = t_{v_1}, \quad \rho_2^4 = t_{v_2},$$

and the cusp filling kills Λ_{tor} and imposes

$$(25) \quad \rho_1 \rho_2 = 1.$$

Proof. By proposition 5.3, attaching N_j kills $\widehat{g}_j^{m_j} t_{-v_j}$. A clockwise meridian in the t_j -disc lifts through $t_j = s_j^{m_j}$ to a path on which s_j turns through $-2\pi/m_j$. Its endpoint is obtained by the deck transformation g_j , because $s_j \circ g_j = e^{-2\pi i/m_j} s_j$. In the coordinates of N_j , the lift of the zero section is $\varsigma_j = -\log(s_j)\Pi v_j/(2\pi i)$. The identity

$$R_j(z)\varsigma_j(z) + \frac{1}{m_j}\Pi(g_j z)v_j = \varsigma_j(g_j z)$$

follows from $\log(s_j \circ g_j) = \log s_j - 2\pi i/m_j$ and $A_j v_j = v_j$. Thus the gluing identifies ρ_j with $\widehat{g}_j$, with the same sign for $j = 1, 2$, and (24) follows.

By proposition 4.5, attaching N_0 kills the normal closure of Λ_{tor} and of the toric meridian. Under proposition 4.4, the zero section is the disc $(x_1, x_2) = (1, 1)$, so its clockwise boundary is the toric meridian and is also the lift ρ_0 . Since $\rho_0 = (\rho_1 \rho_2)^{-1}$, (25) follows. $\square$

Theorem 6.2. *The compact complex threefold X is simply connected.*

Proof. Apply van Kampen's theorem to the open cover by the images of J and slightly thickened cusp and logarithmic pieces. Their intersections are connected torus bundles over annuli, so lemma 6.1 gives the quotient of $\pi_1(J)$ by exactly the stated normal relations. The normal closure of Λ_{tor} under A_1, A_2 is $\ker \gamma$ by proposition 2.1. The surviving image of Λ is therefore cyclic, generated by the image c of $\widehat{\gamma}$, and is central because A_1, A_2 act trivially on the coinvariant quotient. By lemma 6.1, with $x = \rho_1$ and $y = \rho_2$, the group has the presentation

$$\langle c, x, y \mid c \text{ central}, xy = 1, x^3 = c, y^4 = c^{-1} \rangle,$$

because $\gamma(v_1) = 1$ and $\gamma(v_2) = -1$. Hence $y = x^{-1}$ and $c = x^3$, while $y^4 = c^{-1}$ gives $x^{-4} = x^{-3}$. Thus $x = 1$, and then $y = c = 1$. $\square$

7. INTEGRAL COHOMOLOGY

Let F be a smooth fibre, marked so that

$$H_1(F; \mathbb{Z}) = \Lambda, \quad H^1(F; \mathbb{Z}) = V, \quad H^q(F; \mathbb{Z}) = \bigwedge^q V.$$

Let $j : B^\circ \hookrightarrow B$ and let L^q be the local system $R^q f_* \mathbb{Z}|_{B^\circ}$.

7.1. Specialization at the multiple fibres. Put

$$\psi_1 = 2u + w + 3\delta, \quad \psi_2 = u + w + 2\delta.$$

The central torus in the covering model of N_j maps to S_j by an m_j -sheeted covering. Using the smooth marking over Δ_j , identify that torus with a nearby fibre F and denote the covering by $\pi_j : F \rightarrow S_j$.

Proposition 7.1. *The groups $H^q(S_j; \mathbb{Z})$ are torsion-free of ranks*

$$(1, 2, 2, 2, 1),$$

and the pullback from S_j to a nearby torus is injective. Its image in the monodromy invariants has the following indices:

q	0	1	2	3	4
$j = 1$	1	3	1	1	3
$j = 2$	1	4	2	2	4

More precisely,

$$\pi_1^* H^1(S_1) = \langle 3\gamma, \psi_1 \rangle, \quad \pi_2^* H^1(S_2) = \langle 4\gamma, \psi_2 \rangle.$$

The degree-two cokernel at p_2 is generated by the class of $q = uw + 6\gamma\delta$, and the degree-three cokernel at p_2 by the class of γuw .

Proof. The fundamental group of S_j has the presentation

$$\langle \Lambda, \widehat{g}_j \mid \widehat{g}_j \lambda \widehat{g}_j^{-1} = A_j \lambda, \widehat{g}_j^{m_j} = t_{v_j} \rangle.$$

The matrices in proposition 2.1 give

$$(A_1 - I)\Lambda = \{\lambda : \gamma(\lambda) = 0, \psi_1(\lambda) = 0\}, \quad (A_2 - I)\Lambda = \{\lambda : \gamma(\lambda) = 0, \psi_2(\lambda) = 0\}.$$

Indeed the generators of $(A_1 - I)\Lambda$ are $6\widehat{u} - 6\widehat{w} - 2\widehat{\delta}$, $-\widehat{u} - \widehat{w} + \widehat{\delta}$ and $\widehat{u} - 2\widehat{w}$, which generate the displayed kernel; the analogous statement follows from the three generators of $(A_2 - I)\Lambda$. The maps $(\gamma, \psi_j) : \Lambda \rightarrow \mathbb{Z}^2$ are surjective, so $\Lambda/(A_j - I)\Lambda \cong \mathbb{Z}^2$.

After abelianization the relation is

$$m_j \widehat{g}_j = (\gamma(v_j), \psi_j(v_j)).$$

For $v_1 = \varepsilon$ this is $(1, 0)$ and for $v_2 = -\varepsilon'$ it is $(-1, 0)$. The relation vector in $\mathbb{Z}^3$ is therefore primitive, and $H_1(S_j; \mathbb{Z}) \cong \mathbb{Z}^2$ in both cases. Since S_j is a free finite quotient of a four-torus, its Euler characteristic is zero. Thus $b_1 = b_3 = 2$ and $b_2 = 2$. Poincaré duality gives $H^3(S_j; \mathbb{Z}) \cong H_1(S_j; \mathbb{Z}) \cong \mathbb{Z}^2$. In the universal coefficient sequence for H^3 , the torsion group $\text{Ext}(H_2(S_j; \mathbb{Z}), \mathbb{Z})$ injects into this free group; hence $H_2(S_j; \mathbb{Z})$ is torsion-free. Poincaré duality and the universal coefficient theorem now show that every $H^q(S_j; \mathbb{Z})$ is torsion-free, with ranks $(1, 2, 2, 2, 1)$. The cohomological transfer of π_j satisfies $\text{tr} \circ \pi_j^* = m_j$; hence π_j^* is injective in every degree.

A homomorphism $\pi_1(S_j) \rightarrow \mathbb{Z}$ restricting to $\xi \in V^{T_j}$ exists exactly when m_j divides $\xi(v_j)$. Since

$$V^{T_1} = \langle \gamma, \psi_1 \rangle, \quad V^{T_2} = \langle \gamma, \psi_2 \rangle,$$

and $\psi_j(v_j) = 0$, the degree-one images are the displayed lattices.

The invariant degree-two lattices are

$$(\bigwedge^2 V)^{T_1} = \langle \gamma \psi_1, q \rangle, \quad (\bigwedge^2 V)^{T_2} = \langle \gamma \psi_2, q \rangle.$$

Their intersection matrices, with values in $\mathbb{Z} \text{ vol}$, are

$$\begin{pmatrix} 0 & 3 \\ 3 & 12 \end{pmatrix}, \quad \begin{pmatrix} 0 & 2 \\ 2 & 12 \end{pmatrix},$$

with determinants -9 and -4 . The intersection form on $H^2(S_j; \mathbb{Z})$ is unimodular. It is even. Indeed, the covering torus has a nowhere-vanishing holomorphic two-form ω_0 . At the central point the linear part of the deck generator is $R_j(z_j)$, so ω_0 is multiplied by $\det R_j(z_j)$. From proposition 3.2 and equation (9),

$$\det R_1(z_1) = -\rho^{-1} = e^{2\pi i/3}, \quad \det R_2(z_2) = i^{-1} = -i.$$

The form ω_0 identifies the pullback of K_{S_j} with the trivial line bundle equipped with the character $\widehat{g}_j \mapsto \det R_j(z_j)^{-1}$. This character has finite order, so K_{S_j} is torsion and hence numerically trivial. Wu's formula gives $x^2 \equiv x \cdot c_1(K_{S_j}) \equiv 0 \pmod{2}$ for every $x \in H^2(S_j; \mathbb{Z})$. Pullback multiplies the intersection form by the covering degree m_j . If k_j is the index of its image in the invariant lattice, determinant comparison gives

$$k_1^2 = \frac{3^2}{9} = 1, \quad k_2^2 = \frac{4^2}{4} = 4.$$

Thus the degree-two indices are 1 and 2. The vector q is not in the image for $j = 2$: otherwise $q = \pi_2^* a$ would give $12 = q^2 = 4a^2$, so $a^2 = 3$, contradicting evenness. Hence its class generates the cokernel.

The degree-three indices follow from the perfect pairing $H^1 \times H^3 \rightarrow \mathbb{Z}$. Direct calculation gives

$$\begin{aligned} (\bigwedge^3 V)^{T_1} &= \langle \gamma u w, -u w \delta + 2 \gamma w \delta + 4 \gamma u \delta \rangle, \\ (\bigwedge^3 V)^{T_2} &= \langle \gamma u w, -u w \delta + 3 \gamma w \delta + 3 \gamma u \delta \rangle. \end{aligned}$$

The determinants of the pairings with the degree-one invariant lattices are 3 and 2. Let $k_j^{(r)}$ be the index of the pullback image in degree r . If bases of the pullback lattices are expressed in

bases of the invariant lattices, the determinants of the two change-of-basis matrices are $k_j^{(1)}$ and $k_j^{(3)}$. On the other hand, pullback multiplies the perfect pairing on S_j by m_j , so the determinant of the restricted rank-two pairing is m_j^2 . Therefore

$$k_j^{(1)}k_j^{(3)} |\det((\bigwedge^1 V)^{T_j} \times (\bigwedge^3 V)^{T_j})| = m_j^2.$$

Using $k_1^{(1)} = 3$, $k_2^{(1)} = 4$ gives $k_1^{(3)} = 1$, $k_2^{(3)} = 2$. Moreover ψ_2 lies in the degree-one pullback image, whereas every pairing between pullback classes in degrees one and three is divisible by 4. Since $\langle \psi_2, \gamma uw \rangle = -2$, the class γuw is not in the degree-three pullback image and therefore generates its order-two cokernel. In degree four pullback is multiplication by the covering degree. $\square$

7.2. Specialization at the cusp.

Theorem 7.2 (Integral invariant cycles at the cusp). *For every q , the specialization map*

$$\mathrm{sp}^q : H^q(W; \mathbb{Z}) \longrightarrow H^q(F; \mathbb{Z})$$

is injective and has image

$$H^q(F; \mathbb{Z})^{T_0}.$$

In particular

$$H^q(W; \mathbb{Z}) \cong \mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z} \quad (q = 0, 1, 2, 3, 4),$$

and all these groups are torsion-free.

7.3. The integral direct-image sheaves. Write

$$R^q = R^q f_* \mathbb{Z}.$$

Proper base change identifies the stalk of R^q at a special point with the cohomology of the corresponding fibre. At p_0 , the map to the nearby invariant lattice is the specialization map of theorem 7.2. At p_j , the local tube retracts to S_j , and the map to a nearby fibre is the pullback along the covering $F \rightarrow S_j$. These maps are injective by proposition 7.1 and theorem 7.2; hence R^q is the subsheaf of $j_* L^q$ determined by the three displayed stalk lattices.

Lemma 7.3 (Cohomology of the extended local systems). *Let L be a local system of free finitely generated abelian groups on B° , with stalk M and monodromy group G .*

(i) *There is a natural isomorphism*

$$(26) \quad H^2(B, j_* L) \cong M_G.$$

It is compatible with pairings of local systems: under (26), cup product by an invariant element is the induced action on coinvariants.

(ii) *Let t_1, t_2 be the clockwise generators of $\pi_1(B^\circ)$, let $t_0 = (t_1 t_2)^{-1}$, and write T_i for their actions on M . Then*

$$(27) \quad H^1(B, j_* L) \cong \frac{(T_1 - I)M \cap (T_0 - I)M}{(T_1 - I)M^{T_2}}.$$

Proof. The quotient $j_* L / j_! L$ is supported at the three punctures and has no cohomology in degree two. Therefore

$$H^2(B, j_* L) \cong H_c^2(B^\circ, L) \cong H_0(B^\circ, L) \cong M_G$$

by Poincaré duality with local coefficients [3, Ch. V]. Naturality of cap product gives compatibility with coefficient pairings.

A class in $H^1(B, j_* L)$ may be represented by an inhomogeneous group cocycle on $\pi_1(B^\circ)$. Extension across the puncture p_i means that its restriction to the local cyclic subgroup is a coboundary; this is equivalent to the condition $c(t_i) \in (T_i - I)M$. Thus the relevant classes are precisely the parabolic cocycles. Subtracting a coboundary, arrange $c(t_2) = 0$; the remaining coboundaries are those defined by elements of M^{T_2} . Put $x = c(t_1)$. The cocycle identity and $t_0 = (t_1 t_2)^{-1}$ give $c(t_0) = -T_0 x$. Hence the parabolic conditions are $x \in (T_1 - I)M \cap (T_0 - I)M$, and the residual coboundaries change x by $(T_1 - I)M^{T_2}$. This proves (27). $\square$

Proposition 7.4. *The quotient j_*L^q/R^q is a skyscraper sheaf with stalks*

q	p_0	p_1	p_2
1	0	$\mathbb{Z}/3$	$\mathbb{Z}/4$
2	0	0	$\mathbb{Z}/2$
3	0	0	$\mathbb{Z}/2$
4	0	$\mathbb{Z}/3$	$\mathbb{Z}/4$

Consequently

$$(28) \quad H^0(B, R^1) = 12\mathbb{Z}\gamma,$$

$$H^0(B, R^2) = 2\mathbb{Z}q,$$

$$(29) \quad H^0(B, R^3) = 2\mathbb{Z}\gamma uw,$$

$$H^0(B, R^4) = 12\mathbb{Z} \text{ vol}.$$

Furthermore

$$(30) \quad H^2(B, R^1) = \mathbb{Z}[\delta],$$

$$H^2(B, R^2) = \mathbb{Z}[\gamma\delta],$$

$$(31) \quad H^2(B, R^3) = \mathbb{Z}[uw\delta],$$

$$H^2(B, R^4) = \mathbb{Z}[\text{vol}],$$

and

$$(32) \quad H^1(B, R^1) = H^1(B, R^2) = 0.$$

The identifications in degree two are compatible with cup products: an invariant class acts on coinvariants by exterior product.

Proof. The skyscraper table is exactly proposition 7.1 and theorem 7.2. Taking global invariants and imposing the local divisibilities gives (28)–(29), using proposition 2.1(vi).

Since the skyscraper quotient j_*L^q/R^q has no degree-two cohomology, lemma 7.3(i) and the coinvariant calculations of proposition 2.1(vi) give (30)–(31) with the stated compatibility.

The maps from the global invariant lattices to the skyscraper quotients are surjective in degrees one and two: in degree one, γ maps to $(1, 1) \in \mathbb{Z}/3 \oplus \mathbb{Z}/4$, and in degree two, q maps to the generator of $\mathbb{Z}/2$. Hence $H^1(B, R^q) = H^1(B, j_*L^q)$ for $q = 1, 2$. Apply lemma 7.3(ii). For $M = V$,

$$(T_1 - I)V = \langle 2\gamma + u + w, 2\gamma - u \rangle, \quad (T_0 - I)V = \langle \gamma, u \rangle,$$

whose intersection is $\mathbb{Z}(2\gamma - u)$. Since $(T_1 - I)\psi_2 = -(2\gamma - u)$, this equals $(T_1 - I)V^{T_2}$. For $M = \bigwedge^2 V$,

$$(\bigwedge^2 T_0 - I)M = \langle \gamma u, \gamma w + u\delta \rangle,$$

while $(\bigwedge^2 T_1 - I)M$ is generated by

$$\gamma u, \gamma w, 2u\delta + w\delta, uw + u\delta - w\delta - 6\gamma\delta.$$

The intersection is $\mathbb{Z}\gamma u$. Since $\gamma\psi_2$ is fixed by $\bigwedge^2 T_2$ and $(\bigwedge^2 T_1 - I)(\gamma\psi_2) = \gamma u$, this is $(\bigwedge^2 T_1 - I)M^{T_2}$. Formula (27) therefore gives the required vanishing. $\square$

7.4. The Leray differential. Consider the integral Leray spectral sequence

$$E_2^{p,q} = H^p(B, R^q f_* \mathbb{Z}) \implies H^{p+q}(X; \mathbb{Z}).$$

Since B has real dimension two, the only possible nonzero differentials are

$$d_2^{0,q} : E_2^{0,q} \longrightarrow E_2^{2,q-1}.$$

Proposition 7.5. *After choosing the sign of the generator $\omega \in H^2(B; \mathbb{Z})$, one has*

$$(33) \quad d_2(12\gamma) = \omega,$$

$$(34) \quad d_2(2q) = \omega[\delta],$$

$$(35) \quad d_2(2\gamma uw) = \omega[\gamma\delta].$$

In particular all three maps are isomorphisms of infinite cyclic groups.

Proof. The first differential is the obstruction to lifting the global section 12γ of R^1 to a class in $H^1(X; \mathbb{Z})$. We compute it by local lifts. Let $U = D_0 \sqcup D_1 \sqcup D_2$ be a disjoint union of small discs and let V be a neighbourhood of the complementary pair of pants, so that $U \cap V$ is a disjoint union of three annuli. For each of the three discs and for V , the five-term Leray exact sequence contains

$$H^1(f^{-1}(Y); \mathbb{Z}) \longrightarrow H^0(Y, R^1) \longrightarrow H^2(Y; \mathbb{Z}),$$

where Y denotes the corresponding base region. The last group vanishes. Thus a section of R^1 lifts to a degree-one class on the inverse image exactly when it lies in the appropriate local specialization lattice. This holds for 12γ by proposition 7.1 and theorem 7.2. On each annulus the two lifts have the same image in $H^0(R^1)$. The five-term exact sequence for the annulus therefore identifies their difference with the pullback of a unique class in H^1 of the annulus. These three classes form a Čech cocycle, and its image under the Mayer–Vietoris isomorphism

$$H^1(U \cap V; \mathbb{Z}) / \text{im } H^1(V; \mathbb{Z}) \xrightarrow{\sim} H^2(B; \mathbb{Z})$$

is exactly $d_2(12\gamma)$. Evaluating a difference on the oriented annular meridian identifies $H^1(U \cap V; \mathbb{Z})$ with $\mathbb{Z}^3$; the image of $H^1(V; \mathbb{Z})$ is the sum-zero sublattice, so the class of (n_0, n_1, n_2) is $\pm(n_0 + n_1 + n_2)\omega$.

Over D_0 , choose the lift to have value zero on the toric meridian, which is null-homotopic in N_0 . Over D_j , if Θ_j is a lift, the relation $\hat{g}_j^{m_j} = t_{v_j}$ gives

$$m_j \Theta_j(\hat{g}_j) = 12\gamma(v_j).$$

Thus the meridian values of the local lifts are

$$4 \quad \text{at } p_1, \quad -3 \quad \text{at } p_2, \quad 0 \quad \text{at } p_0.$$

Let the lift over V have values a_1, a_2 on the first two meridians. The product relation for the three clockwise boundary meridians gives value $-a_1 - a_2$ at the cusp. The three differences between the disc lifts and the V -lift are therefore

$$4 - a_1, \quad -3 - a_2, \quad a_1 + a_2.$$

Their sum is 1. By the preceding Mayer–Vietoris identification, choosing the sign of ω accordingly proves (33).

Write

$$d_2(2q) = a\omega[\delta], \quad d_2(2\gamma uw) = b\omega[\gamma\delta]$$

for integers a, b . Multiplicativity and the relation

$$(12\gamma)(2q) = 12(2\gamma uw)$$

give, using the graded Leibniz rule with the sign $(-1)^1 = -1$ and the identity $[q] = 12[\gamma\delta]$ in the degree-two coinvariants,

$$12b = 24 - 12a, \quad \text{hence} \quad b = 2 - a.$$

Next $(2q)(2\gamma uw) = 0$ because $H^5(F; \mathbb{Z}) = 0$. Under the coinvariant identifications of lemma 7.3, the two coefficient pairings

$$H^2(B, R^1) \otimes H^0(B, R^3) \longrightarrow H^2(B, R^4), \quad H^0(B, R^2) \otimes H^2(B, R^2) \longrightarrow H^2(B, R^4)$$

have absolute value 2 on the displayed generators: on representatives,

$$\delta \wedge (2\gamma uw) = -2 \text{ vol}, \quad (2q) \wedge (\gamma\delta) = 2 \text{ vol}.$$

The graded Leibniz rule applied to the zero product therefore gives $|a| = |b|$. Since $b = 2 - a$, one has $a^2 = (2 - a)^2$, hence $a = b = 1$. This proves (34)–(35). $\square$

Theorem 7.6. *The integral homology of X is that of S^6 .*

Proof. By propositions 7.4 and 7.5, the spectral sequence gives

$$H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = H^3(X; \mathbb{Z}) = 0.$$

Indeed, in total degree one the only term is the kernel of (33); in degree two the three graded pieces are the cokernel of (33), $H^1(B, R^1)$, and the kernel of (34); in degree three they are the cokernel of (34), $H^1(B, R^2)$, and the kernel of (35).

Since $\pi_1(X) = 1$ by theorem 6.2, one has $H_1(X; \mathbb{Z}) = 0$. The universal coefficient theorem and $H^2(X; \mathbb{Z}) = 0$ imply $\text{Hom}(H_2(X), \mathbb{Z}) = 0$, so the finitely generated group $H_2(X)$ is torsion. The group $\text{Ext}(H_2(X), \mathbb{Z})$ injects into $H^3(X; \mathbb{Z}) = 0$; for a finite abelian group it is isomorphic to the group itself. Hence $H_2(X; \mathbb{Z}) = 0$. Integral Poincaré duality now gives

$$H_3(X; \mathbb{Z}) = H_4(X; \mathbb{Z}) = H_5(X; \mathbb{Z}) = 0.$$

Finally $H_0(X; \mathbb{Z}) = H_6(X; \mathbb{Z}) = \mathbb{Z}$ because X is connected and oriented. $\square$

8. RECOGNITION OF THE SMOOTH MANIFOLD

Lemma 8.1. *Let M be a closed simply connected n -manifold, $n \geq 2$, with the integral homology of S^n . Then M is homotopy equivalent to S^n .*

Proof. If $k < n$ were the least integer with $\pi_k(M) \neq 0$, the Hurewicz theorem would give $\pi_k(M) \cong H_k(M; \mathbb{Z}) = 0$, a contradiction. Thus $\pi_i(M) = 0$ for $1 \leq i < n$, and the Hurewicz theorem in degree n gives an isomorphism $\pi_n(M) \rightarrow H_n(M; \mathbb{Z}) \cong \mathbb{Z}$. Choose $u : S^n \rightarrow M$ representing a class mapping to a generator. The map u is an isomorphism on integral homology. Since both spaces have the homotopy type of CW complexes and are simply connected, the homology Whitehead theorem implies that u is a homotopy equivalence [6, Ch. 4]. $\square$

Proof of theorem 1.1. By theorems 6.2 and 7.6 and lemma 8.1, the closed smooth manifold underlying X is a homotopy six-sphere. For $n \geq 5$, the group Θ_n consists of the oriented h -cobordism classes of homotopy n -spheres. In dimensions at least five, the smooth h -cobordism theorem makes two such spheres diffeomorphic whenever they represent the same class [8]. Kervaire and Milnor computed $\Theta_6 = 0$ [7]. Hence X is orientation-preservingly diffeomorphic to the standard S^6 . Transporting the complex atlas of X through such a diffeomorphism gives an integrable complex structure on S^6 . $\square$

APPENDIX A. INTEGRAL NEARBY CYCLES AT THE CUSP

Let $\psi = R\psi_{f_0}\mathbb{Z}$ be the nearby-cycle complex on W in the analytic topology [4, Ch. 4]. Proper base change gives

$$\mathbb{H}^*(W, \psi) \cong H^*(F; \mathbb{Z}),$$

and the edge map $H^q(W; \mathbb{Z}) \rightarrow H^q(F; \mathbb{Z})$ is the specialization map.

A.1. The cohomology sheaves of nearby cycles. At a point of W , the map f_0 has one of the local forms

$$z_0, \quad z_0 z_1, \quad z_0 z_1 z_2.$$

Writing $z_i = r_i e^{i\theta_i}$, the radial constraint in the Milnor fibre of $z_0 \cdots z_k = t$ is an open simplex, while the phase constraint $\theta_0 + \cdots + \theta_k = \arg t$ is a k -torus. Thus the three local Milnor fibres are homotopy equivalent to a point, S^1 , and T^2 . Therefore

$$\mathcal{H}^0\psi = \mathbb{Z}_W, \quad \mathcal{H}^1\psi = \mathcal{V}, \quad \mathcal{H}^2\psi = \mathbb{Z}_P \oplus \mathbb{Z}_Q,$$

where P, Q are the two triple points and $\mathcal{V}$ is supported on the double locus $D = D_1 \cup D_2 \cup D_3$. Write $\iota_k : D_k \hookrightarrow W$ for the inclusions.

Number the six sides of the hexagon cyclically by $C_1, \dots, C_6$ and let D_k be the common image of C_k and C_{k+3} , for $k = 1, 2, 3$. The normalization distinguishes the two branches globally along $D_k \setminus \{P, Q\}$, so the vanishing-cycle local system there is trivial. Orient its generator by increasing argument on the branch coming from C_k . At either triple point the three restriction values n_1, n_2, n_3 satisfy

$$n_1 - n_2 + n_3 = 0.$$

Indeed, in the local model $z_0 z_1 z_2 = t$, the three vanishing circles have homology classes $e_0 - e_1$, $e_0 - e_2$, $e_1 - e_2$ in $\{(a_0, a_1, a_2) \in \mathbb{Z}^3 : \sum a_i = 0\}$. The global signs are as follows. The three preimages of P are $C_1 \cap C_2$, $C_3 \cap C_4$, and $C_5 \cap C_6$; call the corresponding normalization branches b_0, b_1, b_2 . Thus the ordered branch pairs defining D_1, D_2, D_3 are $(b_0, b_1), (b_0, b_2), (b_1, b_2)$, giving the three differences above. The preimages of Q are $C_2 \cap C_3$, $C_4 \cap C_5$, and $C_6 \cap C_1$; call the corresponding branches c_0, c_1, c_2 . The ordered pairs are then $(c_2, c_1), (c_0, c_1), (c_0, c_2)$. The corresponding differences are $e_2 - e_1, e_0 - e_1, e_0 - e_2$, and again satisfy $n_1 - n_2 + n_3 = 0$. Hence there is an exact sequence

$$(36) \quad 0 \longrightarrow \mathcal{V} \longrightarrow \bigoplus_{k=1}^3 \iota_{k*} \mathbb{Z}_{D_k} \xrightarrow{d} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0, \quad d(n_1, n_2, n_3) = (n_1 - n_2 + n_3, n_1 - n_2 + n_3).$$

Exactness is stalkwise. Away from P, Q the restriction map is an isomorphism onto the single local branch of the double locus. At a triple point the stalk of $\mathcal{V}$ is $\text{Hom}(\{(a_0, a_1, a_2) \in \mathbb{Z}^3 : \sum a_i = 0\}, \mathbb{Z})$, and the three restriction homomorphisms are evaluation on the three differences above. Their image is exactly the kernel of $n_1 - n_2 + n_3$, and that linear form is primitive, so the cokernel is $\mathbb{Z}$.

A.2. The cohomology of the central fibre. Let $\nu : dP_6 \rightarrow W$ be the normalization. For a locally constant function on the normalization, define its k -th difference to be the restriction to C_k minus the restriction to C_{k+3} , transported to D_k . With this difference map and the map d of (36), the normalization complex is

$$(37) \quad 0 \longrightarrow \mathbb{Z}_W \longrightarrow \nu_* \mathbb{Z}_{dP_6} \longrightarrow \bigoplus_{k=1}^3 \iota_{k*} \mathbb{Z}_{D_k} \xrightarrow{d} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0.$$

This is again exact on stalks. At a smooth point it is $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$; at a general double point it is the augmented cochain complex of an interval, $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow 0$; and at a triple point it is the augmented cochain complex of a two-simplex, $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}^3 \rightarrow \mathbb{Z}^3 \rightarrow \mathbb{Z} \rightarrow 0$. The signs are the same as in (36) because both complexes use the same ordering of the local branches. Since ν is finite, proper base change gives $R^r \nu_* \mathbb{Z} = 0$ for $r > 0$; the same vanishing is immediate for the closed embeddings ι_k and for the point inclusions. Thus the sheaf cohomology of the terms in (37) is the ordinary cohomology of dP_6 , the curves D_k , and the points P, Q .

The map in degree two $H^2(dP_6; \mathbb{Z}) \rightarrow \bigoplus_k H^2(D_k; \mathbb{Z}) = \mathbb{Z}^3$ sends a divisor to the differences of its intersection numbers with opposite sides of the hexagon. Writing dP_6 as the blow-up of $\mathbb{P}^2$ at three points, with basis H, E_1, E_2, E_3 , the six sides are

$$E_1, H - E_1 - E_2, E_2, H - E_2 - E_3, E_3, H - E_1 - E_3.$$

For $L = aH - \sum b_i E_i$, the difference map is

$$L \longmapsto (s, -s, s), \quad s = b_1 + b_2 + b_3 - a.$$

Its image is the saturated subgroup $\mathbb{Z}(1, -1, 1)$. The hypercohomology spectral sequence is explicit. In sheaf-cohomology degree zero, the complex of global sections is

$$\mathbb{Z} \longrightarrow \mathbb{Z}^3 \xrightarrow{(a_1, a_2, a_3) \mapsto (a_1 - a_2 + a_3)(1, 1)} \mathbb{Z}^2,$$

with first map zero. Its cohomology has ranks 1, 2, 1 in complex degrees 0, 1, 2; this is the cohomology of the dual cell complex, a torus with one vertex, three edges and two faces. The degree-two row is

$$H^2(dP_6; \mathbb{Z}) = \mathbb{Z}^4 \longrightarrow \mathbb{Z}^3,$$

with saturated rank-one image, hence kernel $\mathbb{Z}^3$ and cokernel $\mathbb{Z}^2$. The degree-four row is $H^4(dP_6; \mathbb{Z}) = \mathbb{Z}$. The resolution has only three nonzero terms after $\mathbb{Z}_W$, in complex degrees 0, 1, 2. Hence $d_r = 0$ for $r \geq 3$, while the only possible d_2 has target in an odd cohomology row and therefore vanishes. The spectral sequence degenerates at E_2 . All surviving groups are free, so there is no extension torsion. Therefore

$$(38) \quad H^q(W; \mathbb{Z}) = \mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z} \quad (q = 0, 1, 2, 3, 4),$$

with no torsion.

From (36) and $H^*(\mathbb{P}^1; \mathbb{Z}) = (\mathbb{Z}, 0, \mathbb{Z})$ one obtains

$$(39) \quad H^q(W; \mathcal{V}) = \mathbb{Z}^2, \quad \mathbb{Z}, \quad \mathbb{Z}^3, \quad 0 \quad (q = 0, 1, 2, 3).$$

A.3. The nearby-cycle spectral sequence. The spectral sequence

$$E_2^{a,b} = H^a(W; \mathcal{H}^b \psi) \implies H^{a+b}(F; \mathbb{Z})$$

has the following free E_2 -terms, displayed by rank:

$b = 2$	2	0	0	0	0
$b = 1$	2	1	3	0	0
$b = 0$	1	2	4	2	1
	$a = 0$	$a = 1$	$a = 2$	$a = 3$	$a = 4$

Let

$$\begin{aligned} \alpha &= \text{rk}(d_2 : E_2^{0,1} \rightarrow E_2^{2,0}), & \beta &= \text{rk}(d_2 : E_2^{1,1} \rightarrow E_2^{3,0}), \\ \gamma &= \text{rk}(d_2 : E_2^{2,1} \rightarrow E_2^{4,0}), & \delta &= \text{rk}(d_2 : E_2^{0,2} \rightarrow E_2^{2,1}), \\ \epsilon &= \text{rk}(d_3 : E_3^{0,2} \rightarrow E_3^{3,0}). \end{aligned}$$

These are the only possible nonzero differentials. Since the Betti numbers of $F = T^4$ are $(1, 4, 6, 4, 1)$, total-degree bookkeeping gives

$$4 = 2 + (2 - \alpha), \quad 1 = 1 - \gamma,$$

so $\alpha = \gamma = 0$. In total degrees two and three one obtains

$$\begin{aligned} 6 &= (4 - \alpha) + (1 - \beta) + (2 - \delta - \epsilon), \\ 4 &= (2 - \beta - \epsilon) + (3 - \gamma - \delta). \end{aligned}$$

Hence

$$\beta + \delta + \epsilon = 1.$$

Thus specialization is already injective in degrees 0, 1, 2, 4, and exactly one rank-one differential remains among

$$E_2^{1,1} \rightarrow E_2^{3,0}, \quad E_2^{0,2} \rightarrow E_2^{2,1}, \quad E_3^{0,2} \rightarrow E_3^{3,0}.$$

It remains to control specialization in degree three.

Lemma A.1 (Specialization with supports). *Let K be a compact subset of the smooth locus of W . After shrinking the base disc, there are a neighbourhood K' of K in W and an open embedding*

$$\Theta : K' \times \Delta_r \longrightarrow N_0, \quad f_0(\Theta(w, t)) = t, \quad \Theta(w, 0) = w.$$

If a compact group acts on N_0 , preserves f_0 and K , then K' and Θ may be chosen equivariantly. For $K_t = \Theta(K \times \{t\})$, the composite

$$H^q(W, W \setminus K) \longrightarrow H^q(W) \xrightarrow{\text{sp}^q} H^q(F_t)$$

equals, under transport by Θ ,

$$H^q(F_t, F_t \setminus K_t) \longrightarrow H^q(F_t).$$

Proof. Choose a relatively compact neighbourhood of K on which f_0 is a submersion and a Riemannian metric there. If a compact group is present, replace the neighbourhood by its group saturation and average the metric over the group. Let ξ_1, ξ_2 be the horizontal lifts of $\partial/\partial \Re t$ and $\partial/\partial \Im t$. For $t \in \mathbb{C}$ set

$$\xi_t = (\Re t)\xi_1 + (\Im t)\xi_2.$$

Then $df_0(\xi_t) = t$. For $r > 0$ sufficiently small, the time-one flow of ξ_t is defined on a fixed neighbourhood K' of K for all $|t| < r$. Put

$$\Theta(w, t) = \text{Fl}_{\xi_t}^1(w).$$

Along the flow, f_0 has derivative t , so $f_0(\Theta(w, t)) = t$. At $t = 0$ the map is the identity. If $\Theta(w, t) = \Theta(w', t')$, applying f_0 gives $t = t'$, and injectivity of the time-one flow of the fixed vector field ξ_t then gives $w = w'$. The differential of Θ is invertible along $K \times \{0\}$; after shrinking K' and r , it is therefore an open embedding. The construction is equivariant when the metric is.

Let $Z = \Theta(K \times \Delta_r)$. The restriction $\Theta|_{K \times \Delta_r}$ is proper: for a compact subset L of $f_0^{-1}(\Delta_r)$, its inverse image is a closed subset of the compact set $K \times f_0(L)$. Hence Z is closed in $f_0^{-1}(\Delta_r)$. Excision identifies the relative cohomology of $(f_0^{-1}(\Delta_r), f_0^{-1}(\Delta_r) \setminus Z)$ with that of the product pair $(K' \times \Delta_r, (K' \setminus K) \times \Delta_r)$. Restriction to the slices $t = 0$ and $t \neq 0$ is an isomorphism by homotopy invariance. The specialization map is obtained by restricting from a sufficiently small tube to a nearby fibre, so these identifications give the asserted commutative diagram. $\square$

Let $K \cong T^2$ be an orbit of the compact torus $T_f = \Lambda_{\text{tor}} \otimes S^1$ in the open torus orbit of W . The open orbit is $(\mathbb{C}^*)^2$, so the normal bundle of K in the smooth four-manifold W° is trivial. Apply the equivariant part of lemma A.1; then K_t is a T_f -orbit in the nearby torus. The composite

$$H^3(W, W \setminus K) \longrightarrow H^3(W) \xrightarrow{\text{sp}^3} H^3(F)$$

is the Gysin map of the transported orbit $K_t \subset F$. The Thom isomorphism identifies the source with $H^1(K)$.

Under the exponential coordinates of proposition 4.4, the T_f -action on a nearby fibre is translation by $(\Lambda_{\text{tor}} \otimes \mathbb{R})/\Lambda_{\text{tor}}$. Consequently K_t corresponds to the saturated sublattice $\Lambda_{\text{tor}} \subset \Lambda$. Choose a complementary sublattice Λ' , so $F \cong K_t \times T_B$ with $T_B = (\Lambda' \otimes \mathbb{R})/\Lambda'$. Under this product decomposition, the Gysin map is

$$H^1(K_t) \longrightarrow H^3(F), \quad a \longmapsto a \smile [T_B]^*.$$

Its image is a direct summand of the Künneth decomposition and is therefore a saturated rank-two subgroup. The composite through $H^3(W)$ has rank two, while $H^3(W)$ is free of rank two; hence sp^3 is injective. Its image contains the saturated rank-two Gysin image and has the same rank, so the two images coincide. In particular the specialization image is saturated. Hence both possible differentials landing in $E^{3,0}$ vanish: $\beta = \epsilon = 0$. Therefore $\delta = 1$, and the unique rank-one differential is $E_2^{0,2} \rightarrow E_2^{2,1}$.

It follows that specialization is injective in every degree. Its image is saturated. In degree one, $H^1(F)/\text{im}(\text{sp}^1) = E_\infty^{0,1} = \mathbb{Z}^2$ is free. In degree two, the filtration quotient $H^2(F)/\text{im}(\text{sp}^2)$ has successive graded pieces $E_\infty^{1,1} = \mathbb{Z}$ and $E_\infty^{0,2} = \ker(d_2 : \mathbb{Z}^2 \rightarrow \mathbb{Z}^3)$, both free; hence the whole quotient is free. Degree three was just proved, and in degree four the edge map is an isomorphism. Every specialized class is T_0 -invariant. By proposition 2.1(vi), the invariant ranks are 1, 2, 4, 2, 1, exactly the ranks in (38). Since the invariant lattices are saturated, the saturated specialization image of the same rank equals the full invariant lattice. This proves theorem 7.2.

APPENDIX B. INTEGRAL MATRIX CALCULATIONS

All groups are written in the ordered bases introduced in proposition 2.1.

The images on the lattice are

$$\begin{aligned} (A_1 - I)\Lambda &= \langle -\widehat{u} - \widehat{w} + \widehat{\delta}, \widehat{u} - 2\widehat{w} \rangle, \\ (A_2 - I)\Lambda &= \langle -\widehat{u} + \widehat{w}, -\widehat{u} - \widehat{w} + \widehat{\delta} \rangle. \end{aligned}$$

The relevant 2×2 minors have greatest common divisor one, so both subgroups are saturated. On V ,

$$(T_1 - I)V = \langle 2\gamma + u + w, 2\gamma - u \rangle, \quad (T_2 - I)V = \langle -3\gamma + u, u + w \rangle.$$

Their sum is the saturated lattice $\langle \gamma, u, w \rangle$.

In the degree-two basis $(\gamma u, \gamma w, \gamma \delta, u w, u \delta, w \delta)$, direct application of the exterior-square matrices gives

$$\begin{aligned} (\Lambda^2 T_1 - I) \Lambda^2 V &= \langle \gamma u, \gamma w, 2u\delta + w\delta, uw + u\delta - w\delta - 6\gamma\delta \rangle, \\ (\Lambda^2 T_0 - I) \Lambda^2 V &= \langle \gamma u, \gamma w + u\delta \rangle. \end{aligned}$$

Together, the images of $\bigwedge^2 T_1 - I$ and $\bigwedge^2 T_2 - I$ generate

$$\langle \gamma u, \gamma w, u\delta, w\delta, uw - 6\gamma\delta \rangle.$$

Adjoining $\gamma\delta$ to these five generators gives a basis of $\bigwedge^2 V$; hence this rank-five lattice is saturated. This proves $(\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta]$ and $[uw] = 6[\gamma\delta]$. Solving the kernel equations gives

$$(\bigwedge^2 V)^{T_0} = \langle \gamma u, \gamma\delta, uw, \gamma w - u\delta \rangle,$$

$$(\bigwedge^2 V)^{T_j} = \langle \gamma\psi_j, q \rangle.$$

The intersection of the two lattices for $j = 1, 2$ is $\mathbb{Z}q$: comparison of the γw - and γu -coefficients forces the coefficient of $\gamma\psi_j$ to vanish. Thus $(\bigwedge^2 V)^G = \mathbb{Z}q$. The products are

$$(\gamma\psi_1)q = 3 \text{ vol}, \quad (\gamma\psi_2)q = 2 \text{ vol}, \quad q^2 = 12 \text{ vol}.$$

The invariant generator $\text{vol} \in \bigwedge^4 V$ identifies $\bigwedge^3 V$ equivariantly with the contragredient lattice Λ . Hence $(\bigwedge^3 V)^G = \mathbb{Z}\gamma uw$ and $(\bigwedge^3 V)_G = \mathbb{Z}[uw\delta]$, with no torsion, by proposition 2.1(v). Directly,

$$(\bigwedge^3 V)^{T_0} = \langle \gamma uw, \gamma u\delta \rangle,$$

$$(\bigwedge^3 V)^{T_1} = \langle \gamma uw, -uw\delta + 2\gamma w\delta + 4\gamma u\delta \rangle,$$

$$(\bigwedge^3 V)^{T_2} = \langle \gamma uw, -uw\delta + 3\gamma w\delta + 3\gamma u\delta \rangle.$$

Pairing these bases with (γ, ψ_1) and (γ, ψ_2) gives determinants -3 and -2 , respectively.

REFERENCES

- [1] An AI model internal to Anthropic, *A compact complex threefold fibred by tori over the projective line, and the six-sphere*, manuscript communicated by Levent Alpoge, 2026, available at <https://alpo.ge/s6.pdf>.
- [2] A. F. Beardon, *The Geometry of Discrete Groups*, Graduate Texts in Mathematics 91, Springer, 1983.
- [3] G. E. Bredon, *Sheaf Theory*, 2nd ed., Graduate Texts in Mathematics 170, Springer, 1997.
- [4] A. Dimca, *Sheaves in Topology*, Universitext, Springer, 2004.
- [5] W. Fulton, *Introduction to Toric Varieties*, Annals of Mathematics Studies 131, Princeton University Press, 1993.
- [6] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.
- [7] M. Kervaire and J. Milnor, *Groups of homotopy spheres I*, Ann. of Math. 77 (1963), 504–537.
- [8] J. Milnor, *Lectures on the h-cobordism theorem*, Princeton University Press, 1965.
- [9] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, 1973.